

The Impact of AI on Skill Development and Career Progression in Software Engineering

A Master's Thesis submitted for the degree of
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supervised by
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Affidavit

I, **DIJANA BAKAJAC**, hereby declare

1. that I am the sole author of the present Master's Thesis, "THE IMPACT OF AI ON SKILL DEVELOPMENT AND CAREER PROGRESSION IN SOFTWARE ENGINEERING", 74 pages, bound, and that I have not used any source or tool other than those referenced or any other illicit aid or tool, and
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Abstract

This study examines the impact of AI adoption on skill requirements and career trajectories within the software engineering field. Through a mixed-methods survey of 41 professionals, the study combines quantitative analysis with qualitative findings to capture both measurable patterns and in-depth insights from professionals from the industry. The quantitative findings show that many agree that AI improves productivity and requires constant skill development. However, findings also point out that there is a notable lack of structured support from organizations for AI training and governance. The qualitative analysis discovered six main themes: (1) Acceleration of Daily Work due to the automation of routine tasks; (2) Shifting Relevance of Technical Skills, with less focus on memorizing syntax and manual tasks; (3) Rise of Meta-Skills like system design, critical evaluation of AI outputs and collaboration; (4) Gaps in Organizational Support; (5) Tensions Around Career Progression for junior roles; and (6) Polarity of Perceived Risks and Opportunities between optimism for growth and concerns about deskilling and job security. Analyzed within the conceptual framework of the Dynamic Capabilities Framework, Triple-Loop Model and the Skill-Biased Technological Change, the findings imply that individuals adapt rather quickly to AI, while organizations often lag. An additional study finding is that AI tends to reward flexible professionals with strong meta-skills, which may further widen existing skill gaps. The study recommends that professionals focus on improving their meta-skills. They need to keep their technical knowledge current and stay up-to-date on AI. While organizations ought to create AI adoption frameworks, invest in organized skill development, and follow ethical AI practices. It concludes that lasting success will require both individuals and organizations to adapt together to benefit from AI while reducing its risks.

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1 Introduction

1.1 Background and Motivation

Artificial Intelligence (AI) is the buzzword of the decade, if not a century, and as many state, the cause of the Fourth Industrial Revolution. With no longer being an abstract concept seen in movies and envisioned as part of a distant future, it became a core driver of transformation across industries, making it one of the most prominent topics in the global business discourse today. One of the industries that was first affected by AI prominence is software engineering, where AI has quickly transitioned from an experimental innovation to an essential tool used in engineering teams. Tools like GitHub Copilot and ChatGPT have become mainstream and are used daily by developers in producing code. This marks a big shift compared to the traditional ways software systems were made.

The influence of AI goes far beyond the software engineering industry. It changes daily operations in many businesses, e.g., the finance industry, where AI systems handle important and sensitive operations like fraud detection, algorithmic trading, and credit risk analysis. Healthcare is one of the most regulated fields, but it still uses AI to improve the accuracy of diagnoses and medical imaging (for example, in radiology). AI is also used in manufacturing and logistics for things like predictive maintenance, demand forecasting, and making the supply chain more efficient. AI is slowly taking over simple, repetitive tasks in all of these fields and helping people make decisions by analyzing data through novel approaches. This implies that professionals now more than ever need to acquire new knowledge and adapt their skill set to stay relevant.

The World Economic Forum (2025) could not be clearer on the urgency of the shift caused by AI, stating that *“on average, workers can expect that two-fifths (39%) of their existing skill sets will be transformed or become outdated over the 2025-2030 period. [...] 85% of employers surveyed plan to prioritize upskilling their workforce.”* There is no need to read between the lines to see that this reflects a clear recognition among global businesses that strategic skill development is crucial for remaining relevant in the ever-changing market.

AI represents a major disruptive innovation, like the Industrial Revolution, the rise of the internet, or even electricity. All of these changes had a big effect on society, the economy, and the way people work and live. Referring to Clayton Christensen's

theory of disruptive innovation (Christensen, 1997), AI for sure falls in the category of disruptive innovation and can be viewed not as an incremental technological advance but as a powerful challenger capable of redefining markets, creating new value networks, and fundamentally altering how businesses operate and compete.

Since innovation management was a major focus of the author's MBA program, it seemed important to explore the topics of organizational adaptation in the face of rapid technology change and disruptive innovation. It suggests that these insights are particularly significant in the context of AI considering the need for businesses to incorporate transformative AI solutions into their people development processes and strategy frameworks, even influencing organizational cultures that must be adhered to. In addition to being strategically important, understanding and effectively navigating this innovation landscape is academically rewarding for all current and future leaders who are expected to guide their organizations and ride the wave of change while staying afloat.

1.2 Why this matters

The speed at which AI is being adopted in software engineering shows that IT companies are making big changes in how they do recruitment, retain their talents, and keep their edge over the competition. The author of this paper, who has a background in software engineering and is now in a leadership position, observes that being able to predict and respond well to changes caused by AI is crucial for both talents (employees) and businesses - and for the author herself, as this ability is essential to remaining relevant in her own career. At present, using AI to increase innovation, productivity, competitiveness, and business outcomes while maintaining a robust, flexible, and future-ready workforce is the top priority for executives in general, not just IT leaders.

Using AI regularly and introducing it across multiple disciplines within one organization gives businesses a lot of strategic benefits and in a vast range as well, like faster product delivery cycles, better code quality, better error detection, more efficient operational workflows etc. It also lets engineers move away from routine tasks and focus on more strategic and even creative ones, like coming up with new solutions, managing cybersecurity threats, designing high-level system architectures, and making sure that software products are in line with the company's strategy.

1.3 Problem Statement

The rise of AI in software engineering is quickly changing the way people obtain and improve their technical skills, as well as the paths they take and the methods they use to develop that knowledge. In the past, as in the majority of other professions, software engineers gained expertise gradually, moving from simple (low-risk) tasks to more difficult ones through consistent hands-on experience and being mentored by more senior engineers (usually through a trial-and-error approach). AI (and the automation it brings) drastically reduces participation in those formative early-career experiences, leaving engineers in the dark about basic concepts. This makes it possible for professionals in the software engineering industry to move up quickly in their careers without gaining enough technical depth and critical judgment.

From an organizational perspective, this transformation disrupts now long-term established workforce development strategies. Existing talent models that are usually built upon clear progression through distinct technical skill sets (which are prescribed step by step) no longer adequately reflect the realities dictated by current widespread automation. Organizations now face the pressing challenge of creating new frameworks that support their employee growth and effectively integrate emerging competencies, such as AI oversight and understanding how AI works, responsible technology usage, interpretability, ensuring that its outputs can be explained, and cross-functional integration into their career and talent management structures.

Consequently, this disjunction between traditional career models and the evolving skill landscape driven by AI leaves both engineers and organizational leaders uncertain about future expectations. It is vital to address this structural gap so that software professionals are equipped with the right skills, career advancement continues to remain meaningful, and companies can keep their ability to innovate and maintain technical excellence.

1.4 Research Gap

Research in academia and industry regarding AI in software engineering primarily emphasizes enhancements in technical performance and increases in productivity (McKinsey & Company, 2023). These insights are useful, but they don't often discuss how automation affects career growth, role changes, and long-term job security. Larger studies like the ones done by the World Economic Forum, discuss global trends in reskilling and the transformation of core competencies (World Economic

Forum, 2025). But these studies usually don't go into details and stay at a high level, and don't look at the specific problems that software engineers face.

In addition, there is a very popular belief, particularly in the media and in some industry statements, that software engineers will be the first ones to be replaced by AI. While this perspective attracts a lot of attention on social media and raises concern for some professionals in the field, empirical research points to a more complicated picture: AI is more likely to change workflows and skill requirements than to completely replace engineering jobs. There hasn't been enough research done on how these changes will affect organizational strategies, learning methods and curves, and career progression frameworks for developing engineering talent in an AI world that is ahead of us.

1.5 Research Objectives

The objectives of this thesis are to:

- Analyze how AI adoption is changing technical and soft skill requirements in software engineering.
- Explore perceptions of AI's impact on careers among professionals at different experience levels.
- Identify how organizations are responding to AI adoption in supporting professional development.

1.6 Research Questions

Main question:

How does AI adoption in software engineering influence required skillsets and reshape career progression?

Supporting questions:

1. Which technical skills are becoming less relevant, and which are emerging as essential?
2. How do software engineers perceive the impact of AI on career opportunities and risks?
3. What approaches are organizations taking to address these challenges?

1.7 Significance of the Study

This research paper makes important contributions to both theory and practice.

On the academic front, it addresses a significant gap by offering empirical insights on the impact of AI on skill acquisition and retention, professional development, and career paths in software engineering, and by that advancing theoretical understanding of career transitions that are heavily impacted by technology.

In real life and practice, it gives managers and organizational decision-makers, human capital professionals, and software engineers strategic value by finding (actionable) ways to adapt skill demands imposed by AI. Insights gathered in this paper should directly help with workforce planning, talent management, retention strategies, and career development, which will make the organization more flexible, keep employees engaged and motivated, and give it a long-term competitive edge in the fast-changing technology landscape.

1.8 Structure of the Thesis

The structure of the thesis is as follows:

- Chapter 1 introduces the research context, along with the study's motivation, research questions, and significance concerning AI impact in software engineering.
- Chapter 2 covers literature review and topics such as the use of AI in software development, trends in automation and abstraction, skill transformation, career progression models, and organizational learning.
- Chapter 3 covers the mixed-methods approach, data collection via a practitioner survey, and the methodology employed for both the quantitative and qualitative analysis.
- Chapter 4 presents empirical results and is organized around six major themes that resulted from the survey data and contains both quantitative findings and a qualitative analysis of practitioner perspectives.
- Chapter 5 interprets the results in the light of the theoretical frameworks and research questions. It gives recommendations to organizations and professionals while offering direction for future research.
- Chapter 6 concludes the thesis by summarizing key findings and presenting actionable recommendations for navigating the software engineering domain in the era of AI.

2 State of the Art and Literature Review

2.1 Introduction

Artificial Intelligence has quite quickly positioned itself from being something considered a niche technology into a main topic of today's business world, impacting business models and job markets. In software engineering, especially, AI has become more than just a productivity tool; it's a force that causes a major shift affecting technical practices, workforce dynamics, organizational strategies and future of the careers.

This chapter analyzes how AI impacts the software engineering discipline. It discusses historic changes and past instances of technological disruption and their effects on career paths, and how organizations respond to new challenges. It also reviews abstraction trends in software engineering (with the emergence of development frameworks) and how they influence career development.

The literature review chapter aims to fulfill the three following goals:

- Mapping the current state of AI in software engineering by identifying key tools, its usage, and how they impact engineering practice.
- Examination of theoretical frameworks and studies that shed light on the connection between technological change and workforce dynamics.
- Noting gaps in current research, especially regarding the specific impacts of AI on skills and career development.

This chapter combines academic literature with industry reports and real-world case studies. Its goal is to provide a detailed understanding of how AI is transforming the software engineering profession.

2.2 Artificial Intelligence: An Overview

AI refers to computational systems capable of performing tasks traditionally requiring human cognition, such as learning, reasoning, and natural language understanding (Russell and Norvig, 2020). The ongoing rise in AI adoption mainly comes from advancements and breakthroughs happening in machine learning and deep learning domains. These technologies use large datasets and powerful computing to enable autonomous decision-making (LeCun, Bengio and Hinton, 2015).

AI systems can be categorized into two main types: Narrow AI and Artificial General Intelligence.

Narrow AI or Weak AI are systems built to do a specific task in a limited domain. Some of the best examples widely known and used by general public are the following:

- ChatGPT - very good at generating text but cannot handle unrelated tasks.
- Google Translate - specialized in machine translation.
- Tesla Autopilot - designed for autonomous driving in controlled settings.
- Siri and Alexa - serve as voice-based virtual assistants.

These systems do better than humans in their specific tasks, but they cannot apply their skills outside their specialized area. However, in their specialized area, they can improve efficiency by increasing productivity and automating tasks. Specifically, they can boost performance by automating routine processes and reducing human error, (Brynjolfsson & McAfee, 2014; Cockburn et al., 2018).

Artificial General Intelligence, or AGI, refers to a theoretical type of AI that can carry out a wide range of cognitive tasks with flexibility similar to human intelligence. Even with major progress in generative models, AGI has not been reached yet. Shane Legg, co-founder of DeepMind, said in 2024 that “practical AGI is *“decades away”* (as cited in Yao, 2024). This highlights the general expert agreement that AGI remains a long-term goal.

2.3 Historical Perspective

The change AI is bringing to software engineering careers builds on a longer history of simplifying technology and automation. In the early days of software development, programmers used low-level languages, which required a deep understanding of topics like memory management and hardware interaction. As software engineering as a discipline advanced, different frameworks and libraries (e.g, Spring, Angular, and Laravel, etc) emerged and made repetitive tasks easier while speeding up development. These tools, in their essence, abstracted many technical details, enabling developers to focus on higher-level design but reducing exposure to foundational programming principles (Sommerville, 2011).

This trend further continued with the growth of Integrated Development Environments, code generators, and, more recently, very popular low-code and no-code platforms like Mendix, MS PowerApps, and Webflow, which allow non-technical users to build

applications, making it more accessible to business professionals and reducing reliance on engineers. Each wave of simplification improved efficiency and reduced the number of bugs, but also led to what many call “black box” systems that operate without developers fully understanding how they work.

This change didn’t just alter how software is built, it also transformed roles in the industry. Previously, an “old-school software engineer” managed all aspects of design and implementation. Now, the field has split into several specific specializations: engineers are now more focused on architecture and system-level design, developers usually work on application logic using pre-built frameworks, and framework or platform experts handle the complexity of these underlying tools. Each of these roles demands different skills and career paths, reflecting a greater division of responsibilities.

The historical trend of simplification that made coding tasks easier and faster has now reached a peak with AI, a tool that not only simplifies but also automates, creating a new level of disruption for career paths.

2.4 AI as a Disruptive Innovation

In 1997, Christensen introduced the idea of disruptive innovation, and AI can certainly be considered its prime example. Disruptive innovation as a concept explains how new technologies first target niche markets before replacing already established products, services, and business models (Christensen, 1997). At first, AI was quite limited to experimental uses in research labs and some narrow specialized fields. Since then, though, its use increased dramatically and it is now being considered essential to many business operations, including software engineering. AI is now being so widely adopted that it is actively changing team structures, coding practices and conventions, and development workflows.

In this study, AI is seen as a disruptive innovation that provides contextual background instead of serving as an analytical framework. This idea places AI within a larger history of transformative technologies, however, it is not directly evaluated in the empirical analysis.

2.5 AI in Software Engineering: Current Applications Across the SDLC

Software engineering is one of the most AI focused fields. It includes applications throughout the entire Software Development Life Cycle (SDLC).

2.5.1 Code Generation and Completion

AI coding tools are now a key part of modern development environments. Platforms like GitHub Copilot and Amazon Q use large language models to generate code suggestions, complete functions automatically, and help with debugging. Adoption has quickly increased; 90% of engineering teams report using AI-based coding tools, and nearly half of the code written by developers in supported languages is now generated by AI (Jellyfish, 2025). These tools not only reduce repetitive tasks but also allow developers to concentrate on more complex problem-solving and design work. However, their effectiveness can differ based on experience level, and their integration needs careful management.

2.5.2 Testing and Quality Assurance

AI is becoming essential to automating software testing and in general, quality assurance practices since it can now create parameterized test cases, anticipate modules that are likely to have defects, dynamically analyze code coverage and it can even perform load tests. These modern AI tools are using machine learning algorithms and natural language processing to interpret specifications and create test scenarios that adapt to ever-changing application behavior and business requirements. Since these solutions have self-healing features, they can automatically modify test scripts in response to UI or API changes, which can significantly lower maintenance costs and bring multiple benefits to the business. However, there are still issues with confirming the dependability of AI-generated test results, having false positives, and guaranteeing contextual accuracy. As a result, human supervision and domain knowledge still remain crucial for preserving test quality and compliance requirements.

2.5.3 DevOps and Deployment Automation

By using advanced features like proactive anomaly detection, resource optimization or build failure prediction, AI enhances continuous integration and continuous delivery (CI/CD) processes within DevOps pipelines and practices. These improvements help

maintain system dependability and reduce unplanned outages. This is especially important in sophisticated large-scale cloud-native environments where real-time monitoring is essential like in finance, retail and telecommunication sectors.

2.5.4 Requirements Engineering

Requirements engineering is usually time-consuming and often ambiguous, which makes it a perfect candidate for AI improvements. Modern tools use natural language processing to examine written requirements for clarity, consistency, and completeness. AI can identify vague terms, recommend changes, and even sort and prioritize requirements based on risk or business value. While its use is still developing compared to coding and testing, its importance is likely to increase in complex projects, especially in regulated industries.

2.5.5 Challenges and Limitations

Despite the numerous benefits observed in previous chapters regarding the use of AI, real-world adoption shows mixed results. Developers see AI as a way to boost productivity. However, studies suggest that in some cases, AI can slow down workflows for experienced engineers because of extra validation steps like reviewing AI output or prompting (Kobie, 2025). This demonstrates the importance of integrating AI carefully and that is crucial to match tools with developer skills and the specific context of each project.

[AI-assisted content generated with ChatGPT (OpenAI, 2025), reviewed by the author].

2.6 Impact on Software Engineering Careers

AI is changing how software engineers build their careers and develop their skills, making it both an opportunity and a challenge. One immediate concern is that AI is taking over simple jobs that junior developers usually do. AI now heavily contributes to tasks like debugging, writing boilerplate code, and manual testing that were once necessary for learning new technical skills. This amplifies the likelihood of having “hollow careers,” all over software engineering industry, where professionals may advance quickly without gaining the technical depth needed for long-term growth, critical thinking, problem solving and flexibility.

As AI tools take over routine coding tasks, junior engineers are losing opportunities to build the foundational knowledge and problem-solving skills necessary for career

advancement. Without these hands-on experiences, their growth may become superficial, focused on managing AI outputs rather than mastering essential technical principles, hence this may lead to the aforementioned “hollow careers.”

On the other hand, AI also gives rise to new hybrid roles that combine software engineering skills with expertise in operations, product strategy, user experience, governance, and others. For instance, AI Product Managers need to translate business needs into AI capabilities hence, they must master both technical language and strategic thinking. The growth of large language models has also led to creation of the role called Prompt Engineer (there is a humorous remark that software engineers will become prompt engineers), who designs effective prompts to guide AI behavior. This role often requires skills like advanced logical reasoning, extensive domain knowledge, and a creative approach to problem-solving. Meanwhile, Human-AI Interaction Designers create intuitive systems that enable meaningful, easy and safe collaboration between people and intelligent systems. One of the roles that will become crucial is AI Regulation & Governance Specialists (although it seems that it is currently not emerging as the other roles mentioned in this chapter). Its job will be to ensure AI systems comply with changing legal standards, ethical guidelines, and internal risk policies. These professionals are expected to work across legal, technical, and organizational lines to encourage responsible AI development.

Emergence of these hybrid roles point out what Daugherty and Wilson (2018) call “fusion skills,” which involve working alongside intelligent systems to enhance human strengths.

These trends demand new career frameworks that require flexibility, interdisciplinary learning, and ongoing skill development beyond just technical areas and expertise.

2.7 Industry Case Studies

The examples from top organizations that are discussed in this chapter are not isolated incidents; in fact, they demonstrate a trend in software engineering toward a collaboration between human and AI. The case studies from Google, ANZ Bank, and others offer a clear perspective on this transition. They illustrate how AI tools are improving productivity, changing the developer's role, and requiring a new set of “fusion skills” to handle this evolving landscape.

2.7.1 Google Engineer Productivity Trial

Using AI tools like code assistants increased task completion speed by 21%, especially for complex tasks, according to a randomized controlled trial involving 96 full-time Google software engineers. A very interesting observation from this trial is that developers who were already proficient in writing code benefited the most from AI support. This trial showed that AI productivity gains heavily depend on context and are yielding best results when combined with human oversight (Paradis et al., 2024).

2.7.2 ANZ Bank + GitHub Copilot Pilot

In a large experiment, 1,000 developers at ANZ Bank participated, and it was expected that they use GitHub Copilot in their daily tasks. Results showed clear improvements in developer efficiency and code quality. Many engineers reported moving from routine coding and repetitive work to higher-level design and problem-solving tasks. This real-world case shows how AI can boost productivity and skill growth in traditional workplaces (Chatterjee et al., 2024).

2.7.3 METR Study on Experienced Developers

A recent study by METR revealed quite a complex outcome: experienced developers working on familiar codebases sometimes showed slower performance when using AI assistants. This was due to the need for extra validation steps in their work and debugging of AI-generated output. Hence, this research shows that there is no one-size-fits-all approach to AI. As AI tools are adopted, critical review skills become even more important (Kobie, 2025).

2.7.4 GitHub Copilot: Early Adoption Benchmark

According to GitHub's internal research conducted during Copilot's launch, developers utilized this tool to write up to 46% of their code in supported programming languages with AI support. Acting as a real-time "AI pair programmer," Copilot helped developers offload routine and repetitive tasks, and as a result, engineers were able to concentrate on more demanding tasks like code design and software architecture. Furthermore, the tool demonstrated the ability to enhance rather than replace human developers by speeding up task completion by up to 55%. This research made it easier to understand how AI can improve routine coding tasks (Dohmke, 2023).

2.7.5 IBM Watson Transition

IBM's Watson platform was initially marketed solely as a healthcare AI, but it was later repositioned as an enterprise automation tool that included compliance monitoring and AI-assisted code review. This shift shows how a business can adapt its AI strategy to meet changing market demands, effectively refocusing its technology to support developer roles with smart compliance and security (IBM, n.d.).

These case studies collectively confirm that AI mainly enhances rather than replaces human work. By handling repetitive or data-heavy tasks, these tools elevate the role of engineers. They also require new skills, such as prompt engineering, critical validation, AI system tuning, and cross-disciplinary collaboration.

2.8 Theoretical Frameworks

2.8.1 Skill-Biased Technological Change

Skill-Biased Technological Change (SBTC) is an economic theory that explains how technological advancements favor skilled labor over low-skilled labor. According to Acemoglu and Autor (Acemoglu and Autor, 2011). SBTC indicates that new technologies help high-skilled workers and replace low-skilled labor. Technology takes over routine tasks that often occur in low-skilled jobs thereby it also raises the value of complex, non-routine tasks. This results in a growing wage gap and a change in the labor market.

The rise of AI signals a new phase of SBTC. It extends its effects beyond manual tasks to many routine cognitive tasks, thus speeding up labor market polarization.

Automation of Cognitive Tasks - AI tools like GitHub Copilot and automated testing solutions change the "task content" of jobs in software engineering. They can effectively handle routine cognitive tasks such as writing boilerplate code and quality assurance. This decreases the demand for workers who specialize in these tasks, making their skills less valuable.

Complementarity and the "AI Skill Premium" - for high-skilled workers, AI acts as a strong complementary tool. It takes care of tedious tasks making room for senior engineers to concentrate on higher-level and non-routine work like system architecture and strategic innovation. This creates an "AI skill premium" for professionals who can use AI effectively and manage AI projects, which further raises their value and the wage gap on the market.

Impact on Labor Market Polarization - AI makes the labor market more polarized. There's an increasing demand for innovators and strategists who can create and manage AI systems, as well as for jobs that are hard for AI to automate like the ones needing non-routine manual skill or interpersonal skills. On the other hand, middle-skilled jobs that involve routine cognitive tasks are becoming more vulnerable to automation.

This changes the skills that individual professionals require and also changes what "expertise" means in fields like software engineering, moving the focus from technical execution to strategic management.

2.8.2 Adapting through Dynamic Capabilities

Due to AI's disruptive nature, organizations are forced to come up with a proper response so they are putting in place strategic measures by focusing on developing a workforce that is more resilient and adaptive.

Their three primary tactics are Upskilling and Reskilling, Agile Workforce Planning and Cross-Disciplinary Integration.

Upskilling and Reskilling - to tackle workforce disruptions caused by AI, organizations are making significant investments (both time and money) in establishing upskilling and reskilling programs. These efforts help ensure that employees can work effectively with intelligent systems as routine roles change or disappear due to automation (Bughin et al., 2018). Employees are concerned about this shift, as a 2024 Gallup poll found that nearly 25% of workers fear their jobs may become outdated because of AI, a notable increase since 2021 (O'Brien & Downie, 2024).

Companies are shifting from traditional, more focused, and specialized training to a "learn-it-all" culture, using micro-learning to promote skill development in areas like prompt engineering and data literacy. For instance, IBM has set up internal AI academies. A recent report from the Business Value Institute concludes that about 40% of the workforce will need to be reskilled in the next three years to keep up with AI integration and the changes it causes (O'Brien & Downie, 2024).

The two main strategies are the following:

- Upskilling: Improving an employee's current skills - good example would be customer service representative learning to use a generative AI chatbot.

- Reskilling: Learning a completely new set of skills for a different job (since the current one might become obsolete), like a manual QA tester learning automation to become a test architect for example.

Agile Workforce Planning - firms are moving away from rigid, long-term talent forecasts and toward more flexible models that meet the fast-changing needs of AI-enabled projects. This agile workforce planning focuses less on filling roles and more on redesigning them. It involves identifying tasks that can be automated and reshaping roles to concentrate on high-value, human-focused activities. Many organizations start these changes with cross-functional so-called "tiger teams" or pilot programs (the expected outcome is usually some type of Proof of Concept). This POC approach helps organizations build internal expertise first and test out the water with new workflows before implementing them more broadly. Mentioned methods mirror McKinsey's recommended practices for strategic workforce planning in the AI era (Gandhi, Durth, & Bérubé, 2025).

Cross-Disciplinary Integration - organizations are removing traditional departmental boundaries and siloes to address very complex ethical and technical challenges posed by AI. By implementing this approach, development teams become more diverse by having a wide range of expertise from different domains. For example, software engineers and AI experts now work alongside domain experts, social scientists, ethicists, and legal professionals as part of one team. Through this collaboration, AI systems are more likely to be efficient, equitable, transparent, and in line with laws and social norms. Organizations can better handle risks like algorithmic bias and data privacy concerns if they incorporate ethical governance frameworks into the development process from the very beginning.

These three strategies are connected parts of a larger organizational change. The Dynamic Capabilities Framework was introduced to show how companies build and renew their skills to respond to fast-changing environments (Teece, Pisano, and Shuen, 1997). This framework highlights that long-term success and competitive advantage in unstable markets depend not only on having valuable resources but also on the ability to adjust, integrate, and reorganize skills and competencies from both inside and outside the company.

The organizational strategies for adapting to AI can be directly mapped to three phases Sensing, Seizing and Transforming (Teece, 2007).

Sensing - the strategic first step is to sense the evolving landscape. This involves actively keeping an eye on the market as well as scouting new technologies (e.g., new AI tools) and identifying the skill gaps and opportunities they create. This phase supports Agile Workforce Planning method where companies predict future needs and expect competitor reactions to shape their talent strategy.

Seizing - once opportunities are sensed, firms must seize them by committing resources and designing new business models. This phase is in real life and practice implemented through Upskilling and Reskilling, which represents a direct investment in new capabilities. The seizing phase also includes Cross-Disciplinary Integration, as organizations make an effort to create new and diverse teams and structures to tap into these opportunities and capitalize on it.

Transforming - the final step of this framework is to change the organization's structure and especially culture to support these new capabilities. This means realigning current skills and investing in new ones to make the changes long-lasting. Through this phase, a culture of continuous learning and responsible AI governance becomes a key part of the company's identity.

Teece (2018) further develops this model by linking dynamic capabilities to business model innovation and strategic renewal. He introduces a well-known structure of sensing opportunities, seizing them through timely investment and coordination, and transforming the organization to sustain adaptability. As shown in Figure 1 (Teece, 2018), these three phases are the strategic foundation for managing disruption in the age of AI.

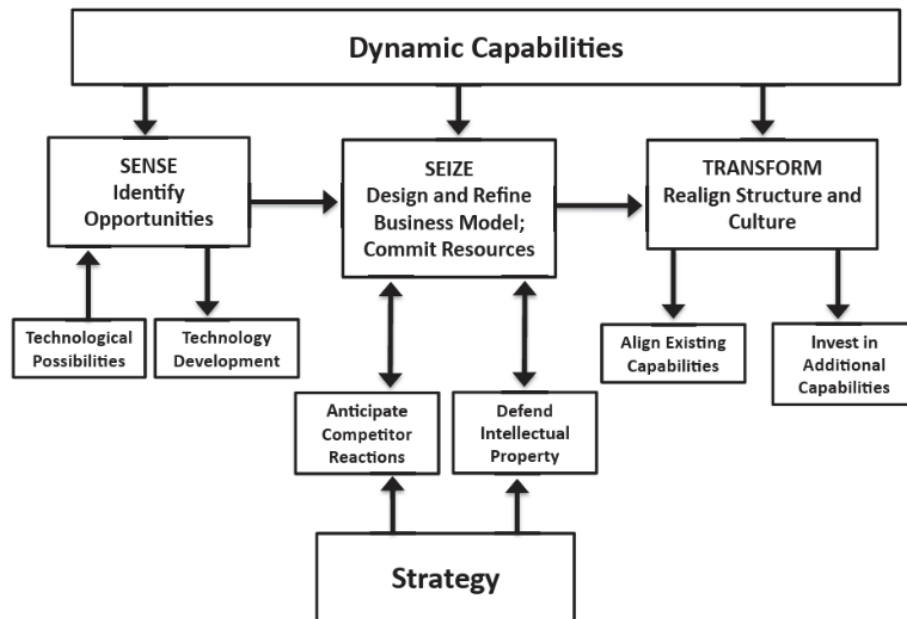


Figure 1: Dynamic capabilities, business models and strategy

By using the Dynamic Capabilities Framework, organizations can shift from a reactive position to a proactive one and have a systematic plan for managing changes driven by AI, ensuring their long-term survival and growth.

2.8.3 Triple-Loop Learning Framework

Incorporating AI into software engineering and understanding the depth of change it causes can be effectively analyzed using the Triple-Loop Learning Framework (Argyris & Schön, 1978; Romme & Van Witteloostuijn, 1999). This model connects innovation and change to different levels of organizational learning and it helps explain how organizations react to AI adoption. Organizations improve processes, as well as transform internal norms and cultural assumptions.

The model identifies three levels of learning: first-order, second-order, and third-order. Each of these levels indicates a deeper stage of change.

First-Order Learning: Incremental Improvements

First-order learning involves small-scale improvements that elevate existing processes without challenging basic assumptions. In software engineering, this is the most immediate effect of AI. For example, tools like GitHub Copilot or AI testing apps help developers write code faster and create test cases more efficiently. The main workflows stay the same. Developers still write and test code, but they do it more

efficiently. AI is used to improve the development process, not fundamentally change it.

Second-Order Learning: Paradigm Transformation

Second-order learning happens when an organization questions and changes the norms that shape its routines. This could entail changing conventional software roles or procedures in light of AI adoption. For example, current manual CI/CD procedures may be replaced by AI-managed pipelines. The engineer's responsibilities for designing, auditing, and maintaining automated systems shift from operator/technician to architect. These changes break away from traditional routines and require a new understanding of responsibilities and expectations.

Third-Order Learning: Mutational Change

Deep cultural change is a component of third-order learning, also known as mutational learning. It demonstrates an organization's capacity for change as well as its ability to challenge its operational guidelines. A business enters this stage when it uses AI to rethink innovation, learning practices, and collaboration in addition to just automating tasks. For example, implementing ethical AI governance, encouraging interdisciplinary roles like AI Product Manager or Human-AI Interaction Designer, and promoting a culture of continuous adaptation are all signs of third-order change. In this phase, organizations view AI as a strategic partner in their evolution.

When analyzing qualitative data from professionals at various career stages, this multi-layered understanding can be of significant value. Junior engineers are usually expected to focus on first-order changes (e.g., task optimization), senior leaders may recognize second- or third-order effects (e.g., changes in accountability, decision-making, and talent strategy). Triple-Loop Learning Framework allows for a structured interpretation of the depth of change within software organizations as they adopt AI.

2.9 Ethical and Societal Implications

Apart from technological and economic shifts, the use of AI in software engineering presents significant ethical and societal challenges. As AI systems more and more start to assist in software development, they address existing issues inherent to engineering practices while also introducing new ones related to AI use. Therefore, it's essential that these issues are discussed and, more importantly, proactively addressed.

Algorithmic Bias and Discrimination: AI systems, especially those trained on large datasets, risk continuing or even worsening the biases present in the training data. For example, a code completion tool trained on a codebase primarily created by one demographic may favor certain coding styles and ignore others, unintentionally setting exclusionary standards. In critical areas like hiring or performance evaluations in engineering, AI-assisted tools could discriminate against certain groups if we do not thoroughly check them for fairness (Floridi et al., 2018). Professionals must make a conscious effort to collect diverse training data, design algorithms focused on fairness, and rigorously evaluate these tools against ethical standards throughout development to mitigate this risk.

Explainability and Accountability: Many AI models, especially deep learning systems, are "black-box" in nature, making it challenging to comprehend how they operate. This can undermine trust. Engineers often struggle to figure out why AI tools suggest significant code changes or point out security holes. This lack of transparency is especially concerning in regulated industries like healthcare and finance, where having proof and an explanation is not only desirable but also required by law. Organizations must establish strong oversight protocols and maintain transparent accountability for AI-generated outputs to ensure that human engineers are ultimately in charge of making decisions.

Job Displacement and Workforce Morale: While many discussions present AI as a tool that improves our work, automating basic tasks like boilerplate coding and routine debugging raises concerns about job security, particularly for junior developers. The fear of being replaceable can hurt morale, slow down acceptance, and create resistance in the workplace. Frey and Osborne (2017) note that anxiety about job loss due to automation continues to be a primary concern in society. This shows how important it is to have clear communication and effective training programs. These programs can help show that AI is a chance for career growth instead of a threat to jobs.

Regulatory and Legal Compliance: The rapid pace of AI development is definitely outstripping the changes in regulatory rules that must keep up. Software engineering now needs to deal with complex legal topics, including data protection (e.g, GDPR), intellectual property, and new regulations like the EU AI Act, which focuses specifically on AI. This creates challenges in making sure that AI-generated code does not unintentionally use copyrighted material, introduce security vulnerabilities, or handle sensitive user data. Engineers must develop skills that blend technical knowledge with

an understanding of legal issues (Russell and Norvig, 2020). The global legal environment is still unclear and chaotic, although the European Union is setting the standard for regulations, and other nations are trying to follow. This situation encourages organizations to adopt flexible compliance strategies that can adjust to evolving rules and unexpected legal changes.

To address these challenges, a high level of proactivity is required, along with embedding responsible AI principles at every stage of the software development process, from design to post-go-live care. For AI to be developed and used effectively, fairly, and transparently, and in alignment with societal values, governance frameworks need to be established both within organizations and at the regulatory level.

[AI-assisted content generated with ChatGPT (OpenAI, 2025), reviewed by the author].

2.10 Research Gap

Existing literature offers a solid foundation regarding AI's technical capabilities and its broader implications in software engineering, and although it seems that especially in the last five years academia is trying to keep up with fast pace of AI development, numerous important areas remain insufficiently explored. This research aims to explore these issues, focusing on individual experiences of industry professionals and organizational responses to changes caused by AI.

Skill Development and Career Progression - productivity metrics like task completion speed or code output volume are often the focus of different studies, particularly in recent years, where year-by-year metrics are compared as AI tools become more sophisticated. On the other hand, not much is known about how automating routine tasks like writing boilerplate code or debugging impacts engineers' ability to develop their skills over time. From the inception of software engineering as a discipline, these routine tasks have been crucial for building technical expertise among professionals. Hence, the threat of "hollow careers," where engineers advance without gaining fundamental skills, has not been studied enough. This research looks at how early-career professionals notice this change and adjust their learning strategies, but also how senior mentors perceive it.

Organizational Talent Strategies - frameworks like Dynamic Capabilities (mentioned earlier in this chapter) promote upskilling and reskilling, but there is little real-world and market insight on how organizations implement and utilize these strategies.

Questions remain about program design, effectiveness, and long-term results. This study looks at how businesses are implementing talent development to stay competitive and up-to-date in the AI era.

Shifts in Professional Identity - while the concept of role redefinition is frequently analyzed in academic discussions, its implications for the personal and professional identities of software engineers are often overlooked. We still lack understanding how professionals feel about AI's impact on their autonomy and career goals. By capturing these viewpoints, this research adds a human-focused element to understanding AI's broader effect.

By addressing these gaps, the study seeks to broaden perspectives and understanding of how AI is changing both the software engineering work and the professionals involved.

[AI-assisted content generated with ChatGPT (OpenAI, 2025), reviewed by the author].

2.11 Conclusion

This chapter brought together theoretical and practical insights about AI's changing role in software engineering. The literature review looked at the historical trend of technological abstraction and showed how AI acts as a disruptive force using frameworks like Skill-Biased Technological Change. It also explored AI's current usage (at least the most prominent one) throughout the software development lifecycle. The chapter examined the significant effects on individual careers, discussing risks of having "hollow careers" all over software engineering practice and the rise of hybrid roles. It considered how organizations respond to these changes through various frameworks. Additionally, it identified important ethical and social challenges that come with this shift. By pointing out the gaps in current research and academic discourse - especially regarding the micro-level effects on skill acquisition, the effectiveness of organizational strategies, and the development of professional identity- this chapter laid the groundwork for the research goals of this study. The next chapters will outline the methodology and research findings aimed at exploring these overlooked areas.

3 Methodical Approach

3.1 Introduction

This chapter outlines the framework established to address the research inquiries presented in Chapter 1, grounded in the theoretical context of Chapter 2. It serves as a guide for the study's research philosophy, design, and implementation. The chapter begins with a summary of the main idea and design of the research. Further, it describes the methods used to collect data, such as creating the survey tool and the sampling plan. Chapter then delves into the two-part way of analyzing data: thematic analysis for qualitative data and descriptive statistics for quantitative data. Lastly, it covers the topic of the study's ethical considerations and limitations. This chapter explains the methodology and the interpretation of the findings presented in the following chapters.

3.2 Research Philosophy and Design

This study uses a mixed-methods, exploratory design that follows a practical research philosophy. Pragmatism focuses on using the most fitting methods to answer research questions, combining both qualitative and quantitative approaches. This flexible design works well for examining the complex and rapidly changing effects of AI on software engineering careers.

3.3 Data Collection

The primary method for data collection was a semi-structured online survey that combined open-ended (qualitative) questions with closed-ended (quantitative) statements. This mixed-methods approach allowed the collection of detailed stories and feedback from participants, while measuring their general sentiment and providing a more comprehensive understanding of the research topic. The survey was available for a two-week period in July 2025.

The survey instrument was developed to directly address the research questions and was organized into four main sections:

Participant Profile: Questions to understand the respondent's professional background, including *Current role in the organization*, *Years of experience in software engineering*, and *What programming languages, frameworks, or tools do you primarily work with?*

AI Adoption and Impact: Questions focused on AI tool usage and its effects on their work, such as *Are you currently using any AI-powered tools in your work?*, *How has the introduction of AI tools changed your daily tasks or responsibilities?*, and open-ended questions about the skills becoming less important or essential to remain competitive.

Organizational and Personal Development: Questions that explored the role of the company and individual learning, including *How do you usually learn new skills?* and *Do you feel your company supports you adequately in acquiring AI-related skills?*

Future Outlook and Perceptions: Questions to gauge perceptions of AI's impact on careers, such as *Has AI influenced your career plans or future opportunities? How?* and *Do you feel AI makes it harder or easier to start a career in software engineering? Why?*

Apart from open ended questions, a series of six statements were also included as closed-ended questions, where participants were expected to respond using a five-point Likert scale (*Strongly Disagree*, *Disagree*, *Neither Agree or Disagree*, *Agree*, *Strongly Agree*). These statements were designed to measure a general consensus on key themes, including:

- [AI improves my productivity]
- [AI makes some of my current skills less relevant]
- [I need to learn new skills to stay competitive]
- [My company provides enough AI-related training]
- [I feel confident adapting to AI-driven changes]
- [AI creates new opportunities for career advancement]

The survey was created and administered using Google Forms. It was distributed via professional communication channels like email and Microsoft Teams, and also less formal channels like WhatsApp, targeting a broad network of industry experts from the professional and private network of the author (friends, colleagues, clients and peers from the university).

3.4 Sample and Participants

Participants were chosen using a purposive sampling strategy. This strategy places more emphasis on selecting cases with rich informational content than on quantity and statistical representativeness; hence, this non-probability approach is ideal for

qualitative research of this type. Professionals with direct experience with the software development lifecycle were the target audience for the inclusion criteria so this method ensured that the data originated from knowledgeable sources.

The final sample consisted of 41 respondents with different professional roles like software engineers, team leads, engineering managers, HR personnel, DevOps specialists, and other positions prevalent in the software engineering field. While this sample size is not big, it is considered adequate for achieving thematic saturation. This is an important factor in exploratory qualitative research, where data collection continues until no new themes or insights arise.

The sample could be considered geographically diverse since it includes participants from Serbia, Austria, Germany, Switzerland, and Portugal, offering a valuable cross-cultural perspective on the adoption of AI in Europe. Additionally, having participants from different organizational setups like software service companies (outsource model), product companies, telecommunications firms, and large corporations provides a solid foundation for examining how AI impacts different types of businesses in diverse ways.

3.5 Data Analysis

Data analysis used a two-part approach to address the qualitative and quantitative data collected.

Qualitative Data Analysis: The open-ended responses were assessed using thematic analysis, following the six-step method outlined by Braun and Clarke (2006). This method was chosen for its flexibility and effectiveness in finding, analyzing, and reporting patterns (themes) within the detailed textual data.

- Familiarization with the data - all survey responses were read and re-read several times to get a complete understanding of the data's content and tone.
- Initial coding - a systematic coding process was conducted, involving a phrase-by-phrase review of the responses. Descriptive codes were created to highlight interesting features of the data, such as "fear of job loss," "need for soft skills," and "lack of company training."
- Searching for themes - codes that were similar were put together into possible themes. This step also involved searching for patterns and links between the codes.

- Reviewing themes - the themes were reviewed and refined in two steps. First, the coded extracts for each theme were reviewed to ensure they made a consistent pattern. Second, the themes were checked against the whole dataset to make sure they accurately represented the overall data.
- Defining and naming themes - the final themes were given names that encapsulated their meaning and were clearly outlined. The main point of each theme was identified to show what it represented.
- Producing the report - direct quotes from survey participants were used to describe and support the themes. The concluding narrative demonstrated how the results complement and build upon previous research by connecting the themes to the theoretical frameworks covered in Chapter 2.

Quantitative Data Analysis: For the six Likert-scale statements, descriptive statistics was used to summarize the data. The responses were interpreted into a numerical scale (for example, 1 = Strongly Disagree, 5 = Strongly Agree), and the frequency distribution, mode, and median were calculated based on the numeric data. This gave an objective overview of the general feelings on specific issues asked in the survey. It also prepared the ground for comparing these results with more in-depth qualitative insights.

[AI-assisted content generated with ChatGPT (OpenAI, 2025), reviewed by the author].

3.6 Use of Generative AI Tools

This thesis was prepared with the support of generative AI tools.

AI assisted with text drafting, summarization, translation and improving academic language. The main tool that was used was OpenAI's ChatGPT (model 5). It helped to lay out arguments, generate different phrasings and with synonym search, translate texts from the author's native language, check grammar errors and improve readability. All sections where AI-generated content was used are marked in the text.

The author carried out all analytical work, interpreted the findings, and integrated theories. The author also critically reviewed results generated by AI to make sure they were correct, original and relevant to the research goals. Generative outputs were treated as draft suggestions, not as final analytical contributions. The author takes full

responsibility for the accuracy, integrity, and originality of all findings presented in this thesis.

3.7 Ethical Considerations

Maintaining ethical integrity was a key focus throughout the research process. The survey was designed to be entirely anonymous, hence no personal information was collected during the process. Participants gave informed consent by agreeing to a clear statement at the beginning of the survey, which explained the purpose of the research, the voluntary nature of their participation, and their right to withdraw at any time. The research followed the ethical guidelines set by TU Wien.

3.8 Limitations

The mixed-methods design of this study is quite good in offering valuable insights, but it has certain limitations. First, the sample size was relatively small and selected intentionally. Participants were chosen based on their relevance and exposure to the topic rather than randomly. This means that the results are based on the experiences and opinions of a particular group of professionals (even though they are spread out across Europe), which limits how much they can be generalized to the broader population of software engineers.

Second, the study relies on self-reported information. Participants shared their perceptions and experiences, however, this may not always accurately reflect their actual behaviors or conclusions. For example, respondents might unintentionally misremember past events (recall bias) or respond in ways they believe are more socially acceptable (social desirability bias). Although the survey was anonymous to reduce these effects, these biases are common in survey-based research, especially if participants come from a private and professional network of the author.

Despite these limitations, the selected methodology aligns well with the objectives of the study. Rather than producing statistically representative results, qualitative research prioritizes depth over breadth and looks for specific insights and patterns. Given the exploratory nature of the subject and the goal of comprehending new trends in a complex and rapidly evolving field like AI, these trade-offs make sense.

3.9 Summary

This chapter explained the methodological framework used to address the study's research questions. By using a described mixed-methods design and a survey that

combines both qualitative and quantitative elements, the study can uncover the layered effects of AI on software engineering careers. The upcoming analysis, which combines thematic analysis for open-ended questions with descriptive statistics for the Likert-scale items, will provide a solid basis for interpreting these findings. The following chapter will present the results of this analysis in detail, organized around key themes and statistical findings that emerged from the data.

4 Research Problem, Working Definition, and Data Analysis

4.1 Introduction

This chapter shares the study's empirical findings based on the theoretical frameworks explained in Chapter 2 and the methods described in Chapter 3. Chapter starts by restating the research problem and the working definition used in the study. Then, it gives a brief overview of the methods and analytical models that guided the research. The rest of the chapter describes the collected data, presents both quantitative and qualitative results, and offers a descriptive discussion of the key trends found.

The main focus of this research is to understand how adopting AI in software engineering affects required skillsets and changes career progression. While AI provides significant opportunities for automation and efficiency improvements, it also disrupts traditional models for career development and skill acquisition, especially during the early stages of professional growth.

The findings in this chapter come from a mixed-methods survey of 41 software engineering professionals from different roles, industries, and locations. The analysis first describes the characteristics of the participant sample, then presents the quantitative results from Likert-scale items, followed by a thematic synthesis of qualitative responses.

4.2 Research Problem and Working Definition

The research problem covered in this thesis arises from the rapid adoption of AI tools in software development. This change is altering the nature of engineering work and the skills needed for career advancement. In the past, software engineers moved up by working on a series of more complex tasks. They gained essential technical skills through hands-on experience and guidance. Now, as AI handles routine coding, testing, and debugging, these important experiences are being lessened or even skipped. This leads to worries about the emergence of “hollow careers.” People might advance without achieving enough technical depth, which could affect the talent pipelines in organizations over the long term.

For this study, the working definition of the research problem is:
“To investigate how the integration of AI tools into software engineering processes

affects the importance of technical and meta-skills, changes the career paths of individuals with different experience levels and impacts organizational strategies for talent development.”

This definition includes three related concepts that are discussed later in the empirical analysis:

- Skill transformation – identifying which technical skills are losing importance and which new skills are becoming essential in AI-supported environments.
- Career progression – exploring how using AI is affecting opportunities, obstacles, and views on career growth.
- Organizational response – assessing how companies are supporting (or not supporting) their workforce in adjusting to these changes.

4.3 Analytical Framework and Methodological Recap

The empirical research uses the mixed-methods approach detailed in Chapter 3. This combines both quantitative and qualitative elements to give a well-rounded view of the research problem.

Data collection method: A semi-structured online survey was created with four sections:

- Participant profile - professional background, years of experience, current role, and main technologies used.
- AI adoption - use of tools, changes in daily tasks, and reported effects on skills.
- Organizational and personal development - learning practices and availability of company support.
- Future outlook and perceptions - expectations about career progression, opportunities and risks.

Quantitative analysis - Six Likert-scale statements measured agreement level on AI's effect on productivity, skill relevance, learning needs, organizational training, adaptability, and career opportunities. Responses were analyzed by using descriptive statistics like frequency distribution, mode, and median.

Qualitative analysis - Open-ended survey responses were assessed using thematic analysis from Braun and Clarke's (2006) six-step approach. This helped identify six

main themes that pointed out how AI is affecting professional skills, career paths, and organizational readiness.

Analytical models - The analysis design was guided by three theoretical frameworks introduced in Chapter 2: the Dynamic Capabilities Framework, Skill-Biased Technological Change, and Triple-Loop Learning Framework. These frameworks provided a foundation for interpreting the results in the critical discussion found in Chapter 5.

The next sections describe the participant sample and present the survey results in both quantitative and qualitative formats.

4.4 Participant Demographics and Context

To provide more context and background for the results of the findings, this section sheds light on the backgrounds of the 41 professionals who participated in the survey. Survey participants included a variety of experience levels and roles within software engineering and related fields.

4.4.1 Years of Experience

As shown in Figure 2, most participants (58.5%) reported having more than nine years of experience in software engineering or related areas. Then 26.8% had between four and eight years of experience, while 14.6% had fewer than three years of experience. This ratio suggests that the survey sample was skewed significantly more toward mid-level and senior-level professionals who had significant exposure to software engineering technologies and overall workplace practices.

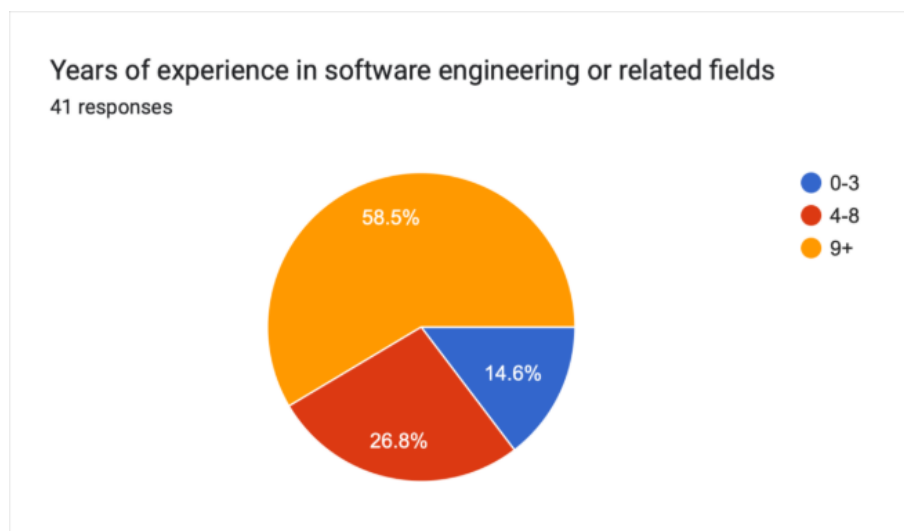


Figure 2: Years of experience in software engineering or related fields

4.4.2 Roles in the Organization

Participants held different roles, showing a diverse perspective on the software industry. Figure 3 shows that almost half of the respondents, 19 out of 41, identified as Software Engineers. Team Leads (5), HR professionals (4), DevOps Engineers (3), Solution Architects (3), and Project Managers (3) were also well represented in the sample. The minority of the sample included CTOs (2), QAs (1), and Business Analysts (1). The sample included both technical experts and management/organizational decision-makers.

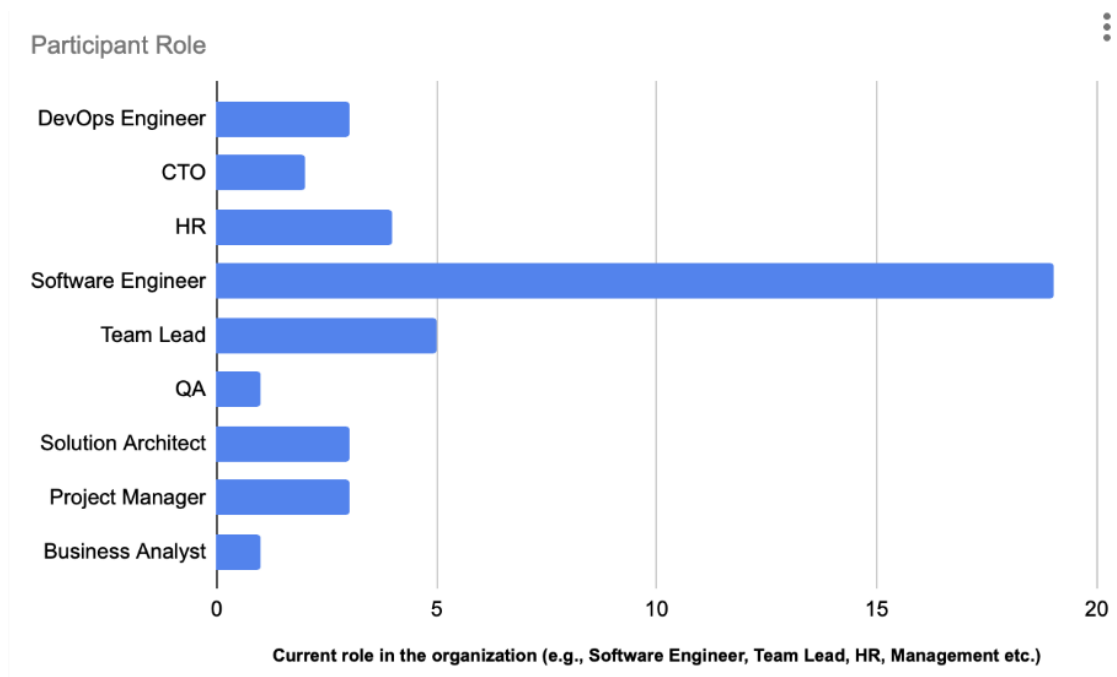


Figure 3: Roles in the organization

Participants represented a range of experience levels and roles, covering different hierarchical and functional positions in software development organizations.

4.4.3 Primary Technologies Used

To better understand the technical context in which respondents use AI tools, participants were asked to list programming languages, frameworks, and tools they mainly use in their daily work. The replies show a mixed (but up to date) tech stack that is quite typical of enterprise-grade and full-stack development environments. Figure 4 provides an overview of the technologies that were most frequently mentioned in the survey responses:

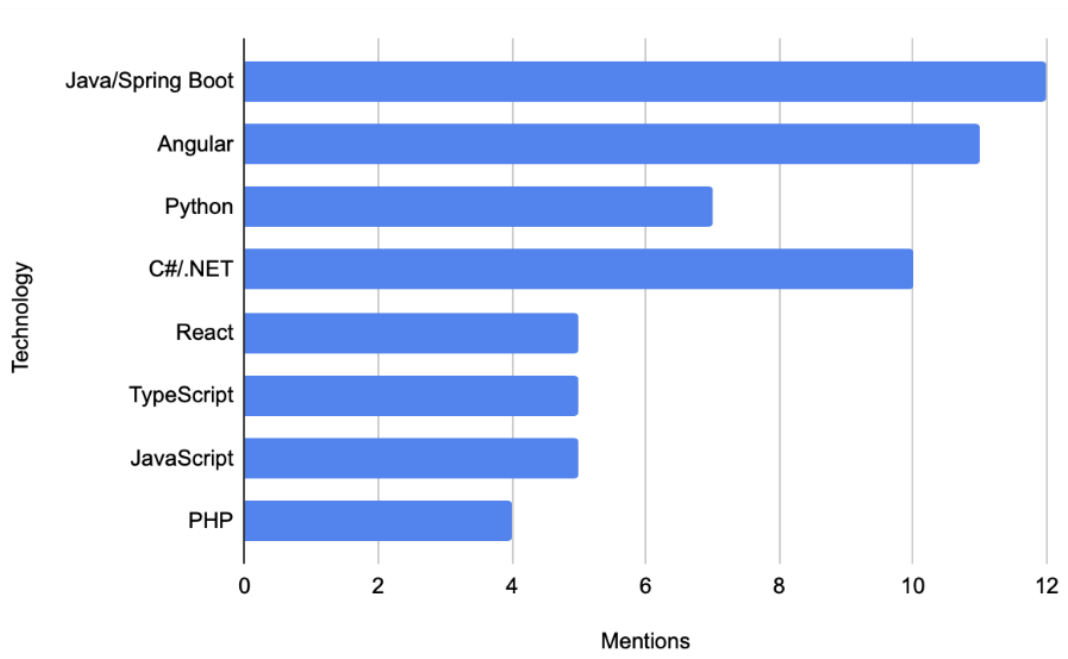


Figure 4: Technology used by survey participants

According to these results, the majority of survey participants use standard frontend and backend technologies mostly in use for web applications and enterprise development. The appearance of Java, .NET, and Spring Boot in survey responses shows strong ties to structured and mainstream software engineering environments. Meanwhile, tools like React, Angular, and TypeScript highlight that respondents have considerable frontend development experience as well.

Understanding this technology landscape is important for interpreting their experiences with AI. Many AI-powered developer tools, such as GitHub Copilot and ChatGPT, are designed to work well (if not primarily) with these languages and environments.

4.5 Quantitative Results

The results of the survey's six Likert-scale statements are shown in this section. The goal was to understand participants' views on how artificial intelligence affects productivity, skills, learning needs, and career opportunities. Respondents rated their agreement on a five-point Likert scale: Strongly Disagree, Disagree, Neither Agree nor Disagree, Agree, and Strongly Agree.

Following the research methods described in Chapter 3, the analysis looks at three descriptive statistical measures suitable for ordinal data: frequency distribution, mode,

and median. These measures help with the clarification of the overall spread and central trends of the responses.

4.5.1 Overview of Responses

The distribution of responses is visualized in Figure 5, which illustrates how participants responded to each statement across the agreement scale.

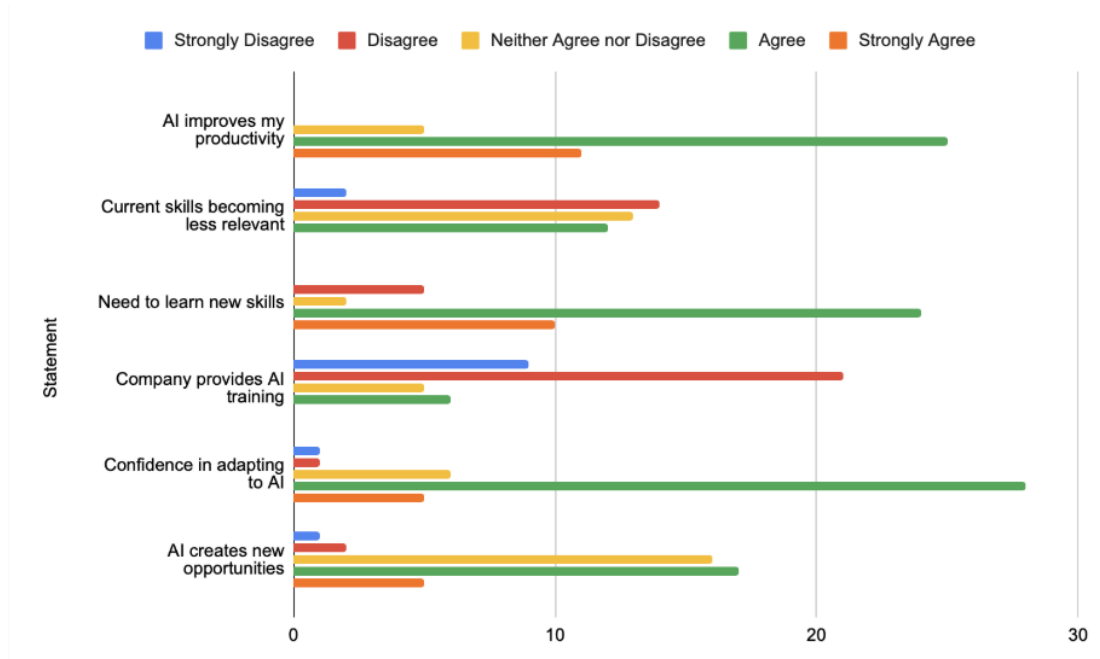


Figure 5: Perceptions of AI's Impact on Productivity, Skills, and Careers

To complement the visual overview, Table 1 presents the exact number of responses (frequencies) for each Likert option. This table is the basis for calculating the mode and median. It also helps support more detailed interpretations throughout the chapter.

Table 1: Frequency Distribution of Likert-Scale Responses

Statement	Strongly Disagree	Disagree	Neither Agree nor Disagree	Agree	Strongly Agree
AI improves my productivity	0	0	5	25	11
Current skills becoming less relevant	2	14	13	12	0
Need to learn new skills	0	5	2	24	10
Company provides AI training	9	21	5	6	0
Confidence in adapting to AI	1	1	6	28	5
AI creates new opportunities	1	2	16	17	5

To highlight central tendencies in the data, Table 2 presents the mode (most frequent response) and median (middle-ranked response) for each statement:

Table 2: Mode and Median of Likert-Scale Responses

Statement	Strongly Disagree	Disagree	Neither Agree nor Disagree	Agree	Strongly Agree
AI improves my productivity	0	0	5	25	11
Current skills becoming less relevant	2	14	13	12	0
Need to learn new skills	0	5	2	24	10
Company provides AI training	9	21	5	6	0
Confidence in adapting to AI	1	1	6	28	5
AI creates new opportunities	1	2	16	17	5

4.5.2 Experience-Based Differences in Perception

To explore how professional seniority level influences their attitudes toward AI, respondents with 0–3 years of experience were compared to those with 9+ years of experience. The sample included 6 junior professionals (0 to 3 years) and 24 senior professionals (9 or more years).

For the statement “*AI improves my productivity*” (Figure 6), junior professionals most often selected Strongly Agree. Senior professionals, on the other hand, most frequently chose Agree.

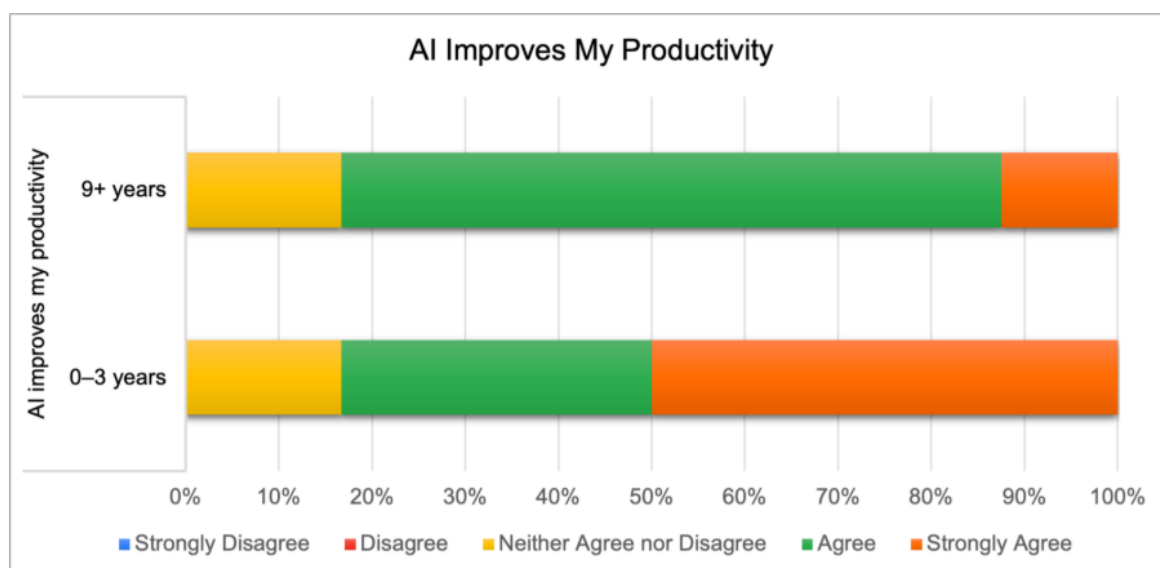


Figure 6: Perception of AI's impact on productivity, comparing responses from professionals with 0–3 years and 9+ years of experience

For the statement “*I feel confident adapting to AI-driven changes*” (Figure 7), senior professionals mainly selected Agree or Strongly Agree. Junior professionals’ responses varied from Disagree to Agree, with none selecting Strongly Agree.

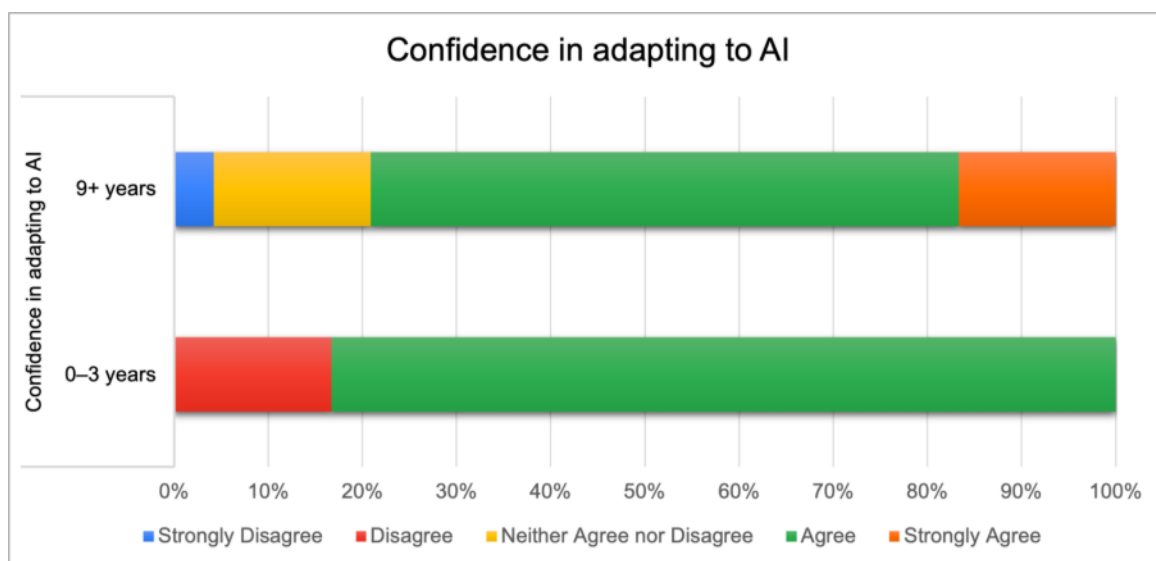


Figure 7: Confidence in adapting to AI-driven changes among professionals with 0–3 years and 9+ years of experience

It is worth noting that the number of senior professionals ($n = 24$) in the sample was much higher than that of junior professionals ($n = 6$), which may affect how generalizable the comparisons are.

These observations show differences in response patterns between the two experience groups. Further interpretation of these differences is presented in Chapter 5.

4.5.3 Summary of Descriptive Statistics

The descriptive statistics for each statement are shown in Table 4.3. Across the six statements, the mode and median values ranged from “Disagree” to “Agree.” These results give an overview of the main trends and differences in participant responses.

Further interpretation of these patterns is presented in Chapter 5.

4.5.4 Summary of Quantitative Results

On the six Likert-scale statements, most participants agreed that AI improves their productivity. They also noted that continuous learning and upskilling are necessary when using AI tools. A majority disagreed that their organizations offer enough AI-related training. Concerns about skills becoming outdated existed but are not

dominant, and most participants felt confident about adjusting to AI. The responses varied based on experience level. Junior professionals (0–3 years) preferred “Strongly Agree” when asked about productivity, while senior professionals (9+ years) more often selected “Agree.” These findings back up the qualitative analysis in Section 4.6.

4.6 Qualitative Data Analysis

To understand how AI is influencing software engineering from the perspective of professionals themselves, a thematic analysis was done based on the qualitative responses from the survey. Following Braun and Clarke’s (2006) six-step methodology, the data were systematically coded, categorized, and refined into a set of meaningful themes. These themes show the experiences, challenges, and opportunities identified by 41 software professionals in different roles and levels of seniority.

The analysis identified six main themes: the acceleration of daily work, the changing importance of technical skills, the emergence of meta-skills, gaps in organizational support, tensions around career advancement, and the contrast of perceived risks and opportunities. Although themes in this section were presented separately for clarity, they are very closely connected.

4.6.1 Theme 1: Acceleration of Daily Work

The most immediate and widely reported effect of using AI in software engineering workflows is the faster completion of routine tasks. Participants said AI tools act as helpers that save time on debugging, coding, and problem-solving.

A mid-level backend developer with 4 to 8 years of experience referred to it as a “*faster way to get to solutions*” by “*automating tedious and repeated tasks*.” A senior developer with more than nine years of experience noted that AI helps with “*less manual boring tasks*.” Another participant mentioned, “*It speeds up development time at least 3 times, as well as skill development*.”

Several participants described using AI tools instead of old methods to find information. One noted: “*I use Git Copilot to speed up writing code and ChatGPT to get ideas for solutions*.” Another stated: “*I use the chat mainly for quickly finding answers*.”

A senior developer with over nine years of experience said: “*It helped me with boring things, like typing long texts. I also use it as help with coding, but not all the time*.”

One participant described AI as a “*sparring partner*” to share ideas and validate concepts. Another noted: “*While AI is great for simple, repetitive tasks, the more complex the problem becomes, the more important it is to write and understand the code manually.*”

Additional perspectives:

"It accelerated on many aspects: mitigated monotonous and repetitive tasks, works as a 'rubber-duck' buddy and helps me find sources and references to documentation for problem solving faster." (Team Lead, 9+ years)

"It changed in a way that some initial research and data collection has been unified and more comprehensive through AI tools... daily tasks' focus shifted from some hard work to more planning, drafting, managing and debugging with 'AI teammate'." (Software Engineer, 9+ years)

"My job became easier." (Software Engineer, 9+ years)

"Extremely for the better. Makes me faster, more productive and provides me with some assistance such as consultation on a daily basis." (Project Manager, 9+ years)

"It sped up a lot of repetitive boring work that does take a lot of time... It's perfect for writing documentation or tickets... Saves a lot of time so that I can do more research, refactor something, maintain code that is more readable and so on." (Software Engineer, 0–3 years)

4.6.2 Theme 2: Shifting Relevance of Technical Skills

Participants regularly highlighted that AI tools are changing the importance of certain technical skills in software engineering. The most common observation is that AI is lessening the importance of repetitive and basic programming tasks like boilerplate coding, memorizing syntax, simple refactoring, and formatting data manually.

Several respondents noted a decrease in the need for skills like memorizing syntax, writing boilerplate code, performing simple refactorings, and manually formatting data.

A mid-level engineer stated that AI handles “*tedious and repetitive tasks such as simple refactorings.*” Another senior professional pointed out that automation now covers “*the repetitive, generic things... like preparing base projects, code blocks for CRUD operations.*”

At the same time, participants underscored that other technical and soft skills are becoming more important. Skills mentioned mainly include architectural thinking, system design, domain knowledge, prompt engineering, and the ability to critically evaluate AI-generated outputs.

A senior developer, with over 9 years of experience, stated the need for the *“ability to critically evaluate and integrate AI outputs,”* while another participant highlighted *“knowing how to use AI to automate certain tasks and help solve the harder ones, without losing critical thinking.”*

Some participants also shared worries about relying too much on AI and the possible decline of basic skills. One junior said, *“You must understand the technology you are working with.”* A senior participant also gave warning, *“If you rely on AI without understanding the fundamentals, you become replaceable. You stop learning, you stop questioning, and your work becomes shallow. The gap will grow between those who use AI to automate thinking and those who use it to amplify their thinking.”*

Additional perspectives:

“Most entry-level skills.” (DevOps Engineer, 9+ years)

“Google search — for that AI is far superior to anything.” (Team Lead, 9+ years)

“Skills like memorizing syntax, writing boilerplate code, or performing repetitive tasks are becoming less important thanks to AI tools.” (Software Engineer, 9+ years)

“Repetitive and manual ones and simple reproduction of knowledge.” (Software Engineer, 9+ years)

“Manual code writing for simple/boilerplate tasks.” (Software Engineer, 4–8 years)

“Many basic data processing tasks, such as manual data entry, formatting, or basic Excel formulas, are becoming less critical due to automation and AI support.” (Project Manager, 0–3 years)

“Fast typing, literary expression, knowledge of grammar, good knowledge of the programming libraries you work with.” (Solution Architect, 9+ years)

4.6.3 Theme 3: The Rise of Meta-Skills

Several participants described the growing importance of meta-skills, which are higher-level cognitive and social abilities. Examples from participants include systems thinking, critical evaluation, communication, and adaptability. These skills are stated

as essential for staying competitive as AI takes over more technical and repetitive tasks.

A CTO explained: *“But what gains importance is architectural thinking, the ability to design scalable, modular systems with flexibility and foresight. It becomes essential to understand how services interact, how data flows, how security and performance play together.”*

A software engineer with 4–8 years of experience noted: *“Before AI you were as good as you can find the answer for your problem on Google. Today you are as good as much as how you can take advantage of AI.”*

A DevOps engineer with 4–8 years of experience highlighted: *“Organizational skills, psychology, communication, analytical thinking.”* Similarly, a mid-level developer stressed: *“Communication. Cooperation with someone more experienced.”*

Other participants highlighted adaptability, curiosity, and critical thinking as essential. A DevOps engineer described adaptability as *“essential, since technologies will constantly change and it is crucial to be able to constantly learn new things.”*

Overall, the responses show a move from just technical skills to meta-skills that combine technical knowledge with wider problem-solving and teamwork abilities.

Additional perspectives:

“Adaptability is essential, since technologies will constantly change and it is crucial to be able to constantly learn new things.” — DevOps engineer, 4–8 years of experience

“Critical thinking, prompt engineering, and the ability to validate and interpret AI-generated content are becoming essential. On the soft skills side, adaptability and digital communication are key to staying competitive in an AI-driven workplace.” — Project manager, 0–3 years of experience

“Knowing how to ask an artificial intelligence question to get the most relevant answer, knowing how to distinguish between good/correct and bad/incorrect/inefficient answer, knowing how to correct/optimize the algorithm suggested by AI so that it does what is expected in the most efficient way.” — Solution architect, 9+ years of experience

4.6.4 Theme 4: Gaps in Organizational Support

The survey data reveals a notable disparity between individual adoption of AI and the lagging response of organizations. While engineers seem to be eager to use AI tools

to improve their productivity and learning many say their organizations do not have a clear and strategic way to integrate AI.

A CTO described the situation as follows: *"Currently provides very limited support when it comes to acquiring AI-related skills or even accessing AI tools in a structured way. Right now, it feels like the potential is acknowledged in theory, but blocked in practice."*

Other participants gave brief but telling responses when asked if their organization supported AI-related training: *"It [the company] doesn't,"* and *"Trying to organise some trainings."*

A senior solution architect explained the limitations of the current approach: *"No. The company is asking us to acquire the skills and giving us the time to do so. Don't get me wrong, it's a valid approach. But then, how can people know when to prioritize such a thing? [...] We need to motivate people to do it, and asking them to do it and giving them something like 2 hours a week may not be the best way to achieve it."*

CTO warned about the risks of this uncoordinated and informal approach: *"At the moment, AI adoption is happening quietly and informally, but not strategically. That needs to change if we want to unlock its full potential in a secure and scalable way."*

Several respondents stated using ChatGPT or Copilot without company resources, using private accounts.

These responses highlight a pattern where AI adoption is driven from the bottom up. It happens without formal structures, secure infrastructure, or consistent governance from the organization.

Additional perspectives:

"No, no one talked to me about AI." (Team Lead, 9+ years)

"No, there is no initiative like paying for us AI tools." (Software Engineer, 9+ years)

"No. Managers in my company are old fashioned." (Software Engineer, 9+ years)

"No. There's no time allocated for upskilling, everyone's firefighting daily tasks." (Software Engineer, 9+ years)

"Our company doesn't organize any courses related to AI skills, and it is up to developers to decide whether to learn about it." (Solution Architect, 9+ years)

"I haven't noticed any support." (Project Manager, 9+ years)

"No and I don't know why." (Software Engineer, 0–3 years)

4.6.5 Theme 5: Tensions Around Career Progression

Survey responses showed a combination of concerns and slight optimism about how AI will impact career growth in software engineering.

A senior solution architect expressed this concern: *“Harder, companies will not hire juniors anymore, as they are often seen as the weakest link in the chain. You will have to study a lot harder because AI has just put the bar higher for the first job.”*

The same respondent added: *“For now, seniors are safe; they are the ones driving it. The issue is that companies thinking in the short term are not investing in new talent under the supposition that AI will be able to do the job they should be doing. This presents a long-term problem: the seniors will eventually leave, and there will be no one to replace them.”*

Other participants described a change in the nature of senior roles.

“Technical execution alone is no longer enough. If machines can generate ideas, write code and optimize processes, then my role shifts toward shaping direction, asking better questions and connecting the dots across disciplines.”

Some respondents stated opportunities AI could create for career growth. A CTO remarked: *“AI creates more opportunities than risks for career development in my field, but only for those who are willing to adapt. The biggest opportunity is leverage.”*

Some respondents stated that while AI might make it easier to start in career it also raises expectations for meaningful contributions. One stated, *“AI makes it easier to start, but harder to truly stand out. It opens the door, but it also raises expectations.”*

A QA professional with 0–3 years of experience connected this shift to fewer mentoring opportunities. *“Their mentoring roles would diminish, due to the shrinking of junior roles. They might be required to take a larger part of tech workload.”*

Not all respondents viewed changes negatively. A CTO described AI as a tool that speeds up progress for those who are willing to adjust: *“With AI, one person can achieve what used to take a team. That opens doors to faster prototyping, broader experimentation and more space for strategic thinking.”*

A software engineer with 9+ years of experience echoed this view: *“Mentors will need to help juniors not only learn coding skills but also develop critical thinking when working with AI-generated solutions.”*

Additional Perspectives:

“Harder at the moment, as it seems to me that the demand for software engineering won’t expand enough in the short term, to compensate for the increase in productivity of currently employed SW engineers due to the integration of AI tools.” (QA, 0–3 years)

“Harder because of false competition. It will make them more significant as someone will have to make decisions and clean up after [ill-conceived] ideas.” (Team Lead, 9+ years)

“AI will shift these roles toward strategic thinking, ethical oversight, and guiding AI adoption.” (Project Manager, 0–3 years)

“The efficient and responsible use of AI might be important here. So it’s mainly about the right decisions where to use AI and how.” (Business Analyst, 9+ years)

4.6.6 Theme 6: The Polarity of Perceived Risks and Opportunities

The survey responses showed that professionals have very different ideas about how AI will affect careers in software engineering in the long run. Some people think AI is a powerful tool that can help people learn and get more done. Others say that there are big risks, especially for new professionals. Some of these risks are higher expectations, fewer chances to learn, and less job security.

Opportunities: Acceleration, Exploration, and Strategic Thinking

Several participants with various levels of seniority described AI as a tool for career growth. It helps them take on more complex tasks, experiment faster, and develop professionally.

“To utilize it as best I can.” (Software Engineer, 9+ years)

“It allowed me to improve myself really quick so I can work on some personal projects.” (Software Engineer, 4–8 years)

“More opportunities definitely. As I’ve said I think of AI as an additional tool in my belt. Same way as the power drill invention helped artisans by allowing them to use their time more effectively.” (Software Engineer, 4–8 years)

“More opportunities as it makes my work more efficient, lets me explore new ideas and application of AI itself to design & build solutions for clients.” (Business Analyst, 9+ years)

“More opportunities to master new skills faster.” (Software Engineer, 9+ years)

Risks: Raised Entry Barriers and Organizational Misalignment

Other respondents stated concerns that AI is making it more difficult to begin a career in software engineering, particularly for junior developers. Some mentioned fewer learning opportunities and an increasing reliance on AI for tasks that used to help entry-level employees build their skills.

“Hype is real, but in reality it has no effect so far. More risks, because people trust it, even though it is not correct, and it says so in every ToS.” (Software Engineer, 9+ years)

“Harder for people who rely on it too much, easier for people who have a solid foundation and are using AI to expand their knowledge.” (Software Engineer, 4–8 years)

“It will be harder, since AI will take over a lot of positions in the IT industry.” (Software Engineer, 4–8 years)

“Harder, because as a junior developer, you lack the necessary experience, preventing you from utilizing the AI the right way. Also the mediors and seniors are significantly faster.” (Software Engineer, 9+ years)

“If there are no juniors and seniors are retiring or getting new job offers, how are you going to get experienced developers?” (Software Engineer, 9+ years)

[AI-assisted content generated with ChatGPT (OpenAI, 2025), reviewed by the author].

4.6.7 Summary of Qualitative Analysis

Across six themes, participants reported a range of experiences with AI in their work. Many noted benefits like quicker task completion, more efficient workflows, and chances to work on more complex or creative projects. However, some pointed out challenges like fewer chances for entry-level learning, increased expectations for output, and limited support from their organizations for AI-related skills. Opinions differed greatly based on role, experience level, and organizational context, showing that AI adoption varies within the profession.

4.7 Summary

This chapter discussed the study's findings, which included numbers from Likert-scale items and insights from open-ended survey responses. The quantitative analysis presented general agreement that AI improves productivity, creates a need for

continuous upskilling and reveals gaps in organizational training. The qualitative analysis expanded on these trends through six themes: acceleration of daily work, shifting relevance of technical skills, the rise of meta-skills, gaps in organizational support, tensions around career progression, and the polarity of perceived risks and opportunities. These results give a clear picture of how AI is changing the skills needed, career paths, and workplace readiness in software engineering. The following chapter expands upon these findings, providing a comprehensive analysis of their significance in relation to the previously examined theoretical frameworks and literature.

5 Discussion

5.1 Introduction

This chapter examines the effects of the findings outlined in Chapter 4. It interprets the quantitative and qualitative results and connects them to the research objectives and questions outlined in Chapter 1. This discussion also looks at how the findings relate to current theoretical frameworks while it also offers practical insights for software engineering professionals and organizations. Finally, it acknowledges the study's limitations and suggests directions for future research.

5.2 Interpretation of Key Patterns – Quantitative

5.2.1 AI Improves Productivity

The majority of participants (both mode and median at Agree) settled on that AI increases productivity. This response pictures a strong belief among professionals that AI can increase efficiency. This supports findings in the literature that AI can significantly speed up software engineering processes by automating repetitive tasks and improving decision-making (Brynjolfsson & McAfee, 2014; Cockburn et al., 2018). These gains are important for the research problem because they let engineers focus more on strategic and creative activities instead of low-level coding.

5.2.2 Current Skills Becoming Less Relevant

Responses varied more here. The most common response was Disagree, but the median was Neither Agree nor Disagree. This shows uncertainty or different experiences with skill changes due to AI.

This fits with the Skill-Biased Technological Change framework (Acemoglu & Autor, 2011). This framework predicts that automation lowers the need for routine skills and raises the need for higher-order skills. This is important for the research problem because it shows how AI-driven change impacts various professionals at different rates. It emphasizes the need for flexibility in career development.

5.2.3 Need to Learn New Skills

Both mode and median were Agree, stressing the shared belief that continuous upskilling is essential in an AI era.

This matches Autor's (2015) point that technological change raises the importance of skills that support automation. It also aligns with Teece's (2007) Dynamic Capabilities framework, which highlights the need to spot and take advantage of new opportunities by building capabilities. This is very important for the research problem because it demonstrates that evolving skills is not optional; it is a strategic requirement for ongoing career growth.

5.2.4 Company Provides AI Training

This item caused the most negative sentiment. With both mode and median at Disagree, the data reveal a clear gap between individual needs for AI trainings and companies having established training programs.

The absence of organized training goes against the proactive "seizing" stage in the Dynamic Capabilities framework (Teece, 2007) and highlights the support gaps within the organization found in the qualitative analysis. For the research problem, this indicates that even though individuals see the need to improve their skills, organizational inertia may hinder the group's capacity to adjust to AI-driven changes.

5.2.5 Confidence in Adapting to AI

Despite the lack of training in the previous bullet, most participants reported feeling capable of adapting to AI. This confidence, reflected in a strong Agree response (mode and median), implies that professionals are resilient and take a proactive behaviour to learning. This resilience connects to the idea of learning agility found in Triple-Loop Learning Framework (Argyris & Schön, 1978; Romme & Van Wittenloostuijn, 1999). In this model, individuals adjust existing practices even if there is no systemic change in the organization. For the research problem, this shows that individual adaptability can help fix some organizational weaknesses. However, without structural support, keeping personal and business outcomes in line over the long term could be at risk.

5.2.6 AI Creates New Opportunities

The central tendency leans positive, with most respondents agreeing that AI is opening new career paths and creating opportunities, however the high number of neutral responses (16) for this question suggests there is room for deeper exploration on this topic.

This shows the dual nature found in the research. AI can enable innovation and disrupt traditional roles (Brynjolfsson & McAfee, 2014; Cockburn et al., 2018). For the research problem, this means that although there is optimism, turning AI's potential into real career benefits relies on personal effort and company strategy.

5.3 Experience-Based Differences in Perception - Interpretation

The comparison between junior professionals (0–3 years of experience) and senior professionals (9+ years of experience) shows apparent differences in how the two groups view AI's effect on productivity and adaptability.

For productivity, early-career professionals were much more likely to choose Strongly Agree as their response. This implies they wholeheartedly adopt AI tools that help simplify their workflows and daily tasks. On the other hand, respondents with over nine years of experience mainly chose Agree, showing a more cautious but still positive view towards AI. This small difference shows a possible gap in expectations. Juniors often view AI tools as essential to their early careers. In contrast, senior professionals compare these tools to more established practices. This pattern fits with Brynjolfsson and McAfee's (2014) observation. They noted that experienced professionals usually assess technological fit based on established methods while newcomers adopt emerging tools without the resistance of long-standing habits.

A more evident gap among different seniorities appeared in responses to the question regarding adaptability. Senior professionals reported very high confidence, with most choosing Agree or Strongly Agree. On the other hand, juniors showed more uncertainty. Some selected Disagree, and none of them chose Strongly Agree. It shows a general lack of self-assurance among juniors in handling the changes that AI brings. This difference likely comes from mileage in the industry, as senior professionals have faced several waves of technological change; they've probably changed several tools, languages, and technologies during their professional life, and as a result, they've learned how to adapt, so AI is just a new technology they would need to master. For juniors, the lack of this "change mileage" creates a vulnerability. They may see fast productivity gains, but they feel less sure about their ability to handle bigger strategic and structural changes that AI could bring. This matches Acemoglu and Autor's (2011) view under the Skill-Biased Technological Change framework. Their view suggests that technological changes often boost the abilities

of skilled workers, which can widen the adaptation gaps between different experience levels.

These results shows that people's perceptions of AI vary based on their experience level and career stage. Juniors seem to be optimistic about AI's instant benefits, but they also feel more vulnerable in terms of strategic adaptation, hence the uncertainty about AI adoption. This points to a need for a proper mentorship and organizational support for professionals at early stages of their career to help bridge this gap.

[AI-assisted content generated with ChatGPT (OpenAI, 2025), reviewed by the author].

5.4 Interpretation of Key Patterns – Qualitative

5.4.1 Theme 1 – Acceleration of Daily Work

Participants largely described AI as a tool to speed up coding, debugging, and information retrieval. More time for significant and intricate tasks is made possible by this efficiency boost. This aligns with SBTC (Acemoglu & Autor, 2011), which suggests that automation shifts value towards tasks that need higher-order thinking. It also supports findings from Brynjolfsson & McAfee (2014) and Cockburn et al. (2018) that AI can boost productivity by improving how humans make decisions and carry them out. For the research problem, this indicates that AI can quickly improve workflow efficiency. It also emphasizes the need to use saved time for strategic and creative problem-solving.

5.4.2 Theme 2 - Shifting Relevance of Technical Skills

Boilerplate coding and repetitive syntax memorization are becoming less significant, according to respondents. The ability to structure systems, integrate various technologies, and assess AI outputs is becoming more and more important. This trend matches SBTC, showing that automation reduces the need for routine skills while increasing the need for adaptable, cognitive skills. It reflects Teece's (2007) idea about adapting key skills to new situations. For the research problem, this states how urgently engineers need to change their technical skills to stay relevant in AI-driven environments.

5.4.3 Theme 3 - Rise of Meta-Skills

Participants consistently emphasized the significance of meta-skills such as critical evaluation, systems thinking, communication, and adaptability. These abilities are essential for overseeing AI processes and guaranteeing high-quality outcomes. This backs up SBTC's claim that workers who can adapt and have a lot of skills are more likely to do well in a changing job market. This shows what Daugherty and Wilson (2018) call "fusion skills." These skills blend human judgment, creativity, and contextual understanding with AI's speed and ability to process data. They enable professionals to work with AI and improve their own contributions through it.

5.4.4 Theme 4 - Gaps in Organizational Support

Many professionals stated that their companies lack structured AI training, policies, or strategic plans for adoption. Individuals are left to learn on their own and use "shadow AI" tools. In the Dynamic Capabilities framework (Teece, 2007), firms should actively invest in building capabilities. This situation goes against the "seizing" and "transforming" phases. In terms of Triple-Loop Learning Framework, this shows a lack of third-order learning—organizations are not rethinking their change processes to effectively integrate AI. For the research problem, this gap explains why individual adaptation is moving ahead of institutional readiness.

5.4.5 Theme 5 - Tensions Around Career Progression

Survey data revealed worries that AI raises the barrier for junior developers, which may reduce mentoring opportunities and create risks for future talent pipelines. Senior professionals reported a shift from hands-on work to strategic oversight and mentoring. This finding connects to SBTC since automation usually increases the need for highly skilled professionals while decreasing entry points for low-skilled workers. For the research problem, this tension indicates a structural challenge in maintaining career advancement paths if foundational roles disappear.

5.4.6 Theme 6 - Polarity of Perceived Risks and Opportunities

Responses were divided between viewing AI as a way or a tool to accelerate career growth and seeing it as a threat to job stability and skill depth. Pessimists pointed to uncertainty, irrational expectations, and organizations not being aligned. In contrast, optimists mentioned quicker learning and taking on more strategic tasks. This shows the two-sided nature of SBTC, it rewards adaptable talent but may leave behind those

who cannot keep up. It also brings to light a challenge related to Dynamic Capabilities. Without the right organizational frameworks to use AI effectively, its benefits are uneven. Regarding the research problem, this gap highlights that AI's effect on careers depends on individual effort and organizational strategy.

5.5 Discussion of Findings in Relation to Research Questions

The study's findings provide a quite comprehensive answer to the main research question by demonstrating that AI adoption has a complex and mixed impact on the software engineering discipline. This section addresses the main research questions, with each supporting question playing a crucial role in the overall response.

5.5.1 Influence on Required Skillsets (Supporting Question 1)

The data clearly state that AI is significantly changing the skillsets needed for software engineers. This is evident in both the quantitative results, where most participants agreed that continuous skill development is essential, and the qualitative Themes 1 (Acceleration of Daily Work) and 2 (Shifting Relevance of Technical Skills). As noted in Theme 1, AI tools are taking over routine and repetitive tasks, resulting in significant productivity gains. This acceleration, however, requires a shift in the relevance of technical skills (Theme 2). Skills like mastering and memorizing syntax, "set and get" coding or manual debugging are becoming less and less important, as AI manages these basic tasks. This doesn't mean that foundational knowledge is no longer needed; instead, how it is applied is evolving by day.

This change highlights the rise of meta-skills (Theme 3), which are becoming crucial for navigating a world enhanced by AI. Cognitive meta-skills like system thinking, architectural design, critical evaluation of AI outputs, and prompt engineering, are now becoming more and more crucial. Engineers are moving from being the main producers of code to becoming managers and validators of AI-generated content. Likewise, interpersonal meta-skills, such as communication, collaboration, and understanding organizational psychology, are becoming more important as effective teamwork with AI and across different functions becomes vital. The findings support broader research indicating a shift toward higher-order cognitive skills and away from routine tasks in an automated setting (e.g., Autor, 2015).

5.5.2 Perceptions of Career Opportunities and Risks (Supporting Question 2)

The study's findings reveal conflicting views on the perceived risks and opportunities of AI's impact on career development. The quantitative data showed this very clearly: most of the answers to "AI creates new opportunities" were positive, but there were also a lot of neutral answers. It was also highlighted in the qualitative themes: 5 (Tensions Around Career Progression) and 6 (Polarity of Perceived Risks and Opportunities). Many professionals, especially those with a proactive mindset, see AI as a tool that speeds up skill development, exploration, and strategic thinking. However, worries about job security and the future of traditional career paths remain significant. This concern is especially strong among early-career professionals as Theme 5 highlights the tensions around career progression. Finding that requires careful consideration is the notion that AI raises the bar for junior developers and may result in a long-term talent gap.

One survey respondent stated this tension quite well with their observation: *"Coding is a language, just like Spanish, English or Portuguese... Take the Portuguese word 'saudade'... That's exactly what happens with code when you only use AI as your translator. It might generate working code, but it won't capture nuance, elegance or intention."*

This "saudade" quote about the depth of the craft points out a big issue. Relying too much on AI as a translator may lead to a shallow understanding of key technical principles. This may create "hollow careers" where engineers on certain (even higher) career ladders lack the skills to innovate or solve complex problems on their own. This is for sure a significant psychological and professional challenge that organizations must urgently address. This fits with the Skill-Biased Technological Change framework (Acemoglu & Autor, 2011). It shows how technology can create high-value opportunities but also push aside those who cannot keep up with changing demands. For the research problem, this means that to make the most of AI's career benefits, focused support is needed for early-career engineers while also safeguarding deep technical skills at every level.

5.5.3 Organizational Approaches to Challenges (Supporting Question 3)

The study found significant gaps in organizational support (Theme 4), revealing that companies are mostly reactive and often unready for the quick individual adoption of

AI. There is a clear need for new skills and career changes, yet formal training programs, clear AI policies, and strategic initiatives are largely absent. Professionals are usually left to their own devices and often have to learn AI-related skills independently, which leads to informal "shadow AI" practices within organizations.

This lack of structured support heightens tensions related to skill development and career progression since many companies claim they want to have "AI First" culture but without clear guidance or significant investments, the burden of adaptation to AI primarily falls on individuals.

From the view of the Dynamic Capabilities framework (Teece, 2007) this shows a failure to progress past the "sensing" stage into proactive "seizing" and "transforming."

For the research problem, this suggests that while individual engineers are adjusting quickly, organizational inertia may hinder overall change. This can create a gap between personal skill growth and the readiness of the organization. To close this gap, organizations need to include AI adoption in their strategic planning, resource distribution, and role design instead of relying on casual, informal experimentation.

[AI-assisted content generated with ChatGPT (OpenAI, 2025), reviewed by the author].

5.6 Implications for Theory

The study's findings contribute to existing theories and frameworks by offering real evidence of AI's ability to disrupt and transform a specific professional domain like software engineering.

5.6.1 Dynamic Capabilities Framework

The observed gaps in organizational support (covered in Theme 4) and the need for strategic AI adoption can be directly related to the Dynamic Capabilities Framework (Teece, 2007). The framework emphasizes a firm's ability to "sense" opportunities, "seize" them through timely investment, and "transform" the organization to stay adaptable. The study's findings suggest that many organizations are currently falling behind in the "sensing" and "seizing" phases when it comes to AI, hence they are, of course, failing in the "transforming" phase as well. Employees are informally adopting AI by identifying (sensing) individual opportunities, which is rarely matched with structured investment (seizing) or systematic reconfiguration of processes, roles, and

strategies (transforming). This situation underscores the urgent need for organizations to build stronger dynamic capabilities. In order to effectively navigate the AI era, organizations need to move from reactive responses to proactive strategic renewal (Teece, 2018).

5.6.2 Triple-Loop Learning Framework

Triple-Loop Learning Framework (Argyris & Schön, 1978; Romme & Van Witteloostuijn, 1999) offers a useful perspective for understanding the extent of change observed.

First-order learning appears in Theme 1 (Acceleration), where AI is used to make small improvements to existing processes, like speeding up code production or debugging.

Second-order learning shows up in Themes 2 (Shifting Skills) and Theme 3 (Meta-Skills). In these themes, engineers start to question and change the norms that guide their routines. They are shifting toward roles that prioritize system design and critical evaluation over routine tasks.

However, the widespread Gaps in Organizational Support (Theme 4) and the Tensions Around Career Progression (Theme 5) indicate that many organizations have not yet reached third-order learning. This deeper change would for sure involve questioning the organization's ability to change itself. It would include integrating ethical AI governance and encouraging a culture of ongoing adaptation throughout the organization. In practice, this might look like including AI-related KPIs in strategic planning. Examples for such KPIs could be "AI-assisted code quality metrics," "reduction in time-to-market for AI-augmented features," or "employee AI literacy scores." It also means redefining engineering roles to encompass AI governance duties like "AI model oversight" and "ethical AI review." Additionally, it requires forming AI ethics committees that guide product development from the very start and incorporating AI insights into business model innovation.

The study shows that while individuals are investing in their learning, organizational learning is mostly stuck at the first or second order which limits a complete adoption of AI.

5.6.3 Skill-Biased Technological Change

The study's empirical findings fit closely to the theory of Skill-Biased Technological Change. This theory implies that technological progress usually raise the demand for

skilled workers while lowering the demand for less-skilled workers (Acemoglu & Autor, 2011). In software engineering, AI seems to be improving the value of meta-skills (Theme 3) like critical thinking, problem definition, and solution design making an "AI skill premium". These skills are usually linked to more experienced or adaptable professionals. On the contrary, the automation of routine, low-level tasks (Theme 1) and the observed issues around career progression (Theme 5) for junior developers point to a possible "hollowing out" of entry-level roles. These roles traditionally served for basic learning experiences. Hence, this matches SBTC's expectation that technology mostly benefits workers with advanced or adaptable skills as well as SBTC pattern of labor market polarization.

5.7 Implications for Practice

The study's findings offer several actionable insights for both individual software engineering professionals and the organizations employing them.

5.7.1 For Professionals

Embrace Meta-Skills - software engineers should work on developing meta-skills like critical thinking, system design, seeing the big picture, and being able to communicate well. As shown in Theme 3, these skills will be crucial for guiding and utilizing AI tools, verifying their outputs, and collaborating effectively within cross-functional teams.

Prioritize Basic Knowledge - even though AI can automate basic tasks, it is important to maintain a strong grasp of core technical fundamentals. As the "*saudade*" quote from one of the respondents suggests, true mastery and the ability to spot subtle errors or create elegant solutions require fluency in coding itself, rather than merely relying on a translator.

Proactive Learning - given that there is currently limited formal support from organizations, individual professionals need to take charge of their ongoing learning and skill development in AI areas. This includes exploring prompt engineering, understanding AI ethics and risks, and constantly trying out new AI tools with a critical eye.

Advocate for Support - due to recognized gap in Theme 4, professionals should be very vocal and express their needs for structured training, access to secure AI tools, and clear career paths to their organizations.

5.7.2 For Organizations

Develop a Strategic AI Adoption Plan - it's evident that companies must urgently transition from informal and ad-hoc AI use to establishing clear frameworks. This change involves setting up explicit policies for AI tool usage and having in mind data security and ethical guidelines for output generated by AI and decision-making. This change will directly tackle the lack of structure in Theme 4 and the “seizing” gap in the Dynamic Capabilities framework.

Focus on Upskilling and Reskilling - organizations need to invest significantly in formal training programs that meet new skill requirements. Simply offering “learning time” in the daily schedule of an employee is not enough. Active incentives, mentorship, and integrating AI literacy into performance frameworks are important.

Redefine Roles and Career Paths - companies should redesign roles to take full advantage of AI and maximize its benefits by focusing on more valuable and human-centered tasks. Clear career paths that include AI-related skills and provide mentorship for younger employees are vital to address concerns about job loss and ensure a strong talent pipeline, thereby mitigating the career progression risks outlined in Theme 5.

Work on a Culture of Continuous Learning and Collaboration - encourage teamwork across different fields by teaming up AI specialists with ethicists, legal experts and domain specialists. Foster a culture that is fail-forward, values continuous learning and shared knowledge.

5.8 Limitations of the Study

While this study offers quite valuable insights, at the same time, it also has several limitations that must be acknowledged. The sample size of 41 participants could be considered diverse in roles and experience but it is still rather small and taken from a specific professional and private network. This may limit how well the findings apply to the wider software engineering industry. The self-reported nature of the survey data could lead to bias, since participants' views might not always match objective facts. Additionally, the cross-sectional design of this research captures only a snapshot in time. Longitudinal studies that cover a longer period would surely provide deeper insights into the changing effects of AI.

5.9 Future Research

Based on the study's results and limitations, several avenues for future research arise and are the following:

Longitudinal Studies - observe how perceptions and skill requirements change over a longer period of time as AI tools become more advanced and crucial element of engineering workflows. One direction could be a particular focus on how early-career vs. senior professionals (Themes 5 and 6) adapt over time and whether mentorship interventions can reduce the observed skill confidence gap.

Comparative Studies - research different industries or regions to understand the differences in AI adoption and its effects on skills and careers.

Organizational Case Studies - conduct detailed case studies of businesses that are known for successfully implementing AI adoption and training programs (and are now yielding results from it) to identify best practices, but also challenges they've faced. To address issues identified in Theme 4, focus should be on how structured support and governance influence both individual skill development and career progression.

Impact on Specific Roles - examine the specific impact of AI on just specific roles from the software engineering discipline like junior developers, QA engineers, or even project/product managers, to have more tailored insights.

Ethical and Psychological Impact - further look into ethical issues and psychological effects of using AI in software development, assessing trust in AI, fear of AI, and information processing demand. Potential focus could be "shadow AI usage" discovered in Theme 4.

[AI-assisted content generated with ChatGPT (OpenAI, 2025), reviewed by the author].

5.10 Conclusion

This study has revealed the significant and multifaceted effects of AI adoption on the software engineering field. The results show that AI is not just a tool for automation, it is also a catalyst for big changes in the skills needed, career paths, and the way organizations function so far. AI certainly has a lot of potential to boost productivity and innovation, however, its usage also raises significant issues like degradation of competencies, anxiety about job security, and a general lack of support from organizations (while also having high expectations due to popular narrative). In the AI

era, the future of software engineering will depend on how quickly and effectively individuals and organizations can adapt to change in order to create an environment that is collaborative but also ethical and continuously learning.

6 Conclusion and Recommendations

6.1 Introduction

This final chapter recapitulates the main findings of the study, restates the answers to the research questions, and highlights the theoretical and practical contributions. It also gives practical recommendations for software engineering professionals and organizations to manage the changing landscape influenced by AI adoption. Lastly, it presents a concluding statement on the wider implications of this research.

6.2 Summary of Findings

This study examined how adopting AI affects skill development and career growth in software engineering. It used a mixed-methods survey of 41 professionals. The findings, which can be found in Chapter 4, showed a complex relationship between new technology and how people adapt to it.

In the quantitative part of the study, the data showed general agreement among participants that AI increases productivity and requires constant skill development. It also highlights a notable lack of official organizational support for AI training and AI governance in general.

On the qualitative side, six connected themes were identified to help illustrate AI's impact on the software engineering field.

Acceleration of Daily Work (Theme 1): A vast number of professionals use AI tools to automate routine tasks or gather information quickly. This increases their efficiency and allows them to focus on more critical and complex responsibilities.

Shifting Relevance of Technical Skills (Theme 2): Traditional low-level coding and memorization are becoming less important. New skills linked to AI interaction, learning, understanding instead of memorizing, and critical evaluation are becoming more valuable.

The Rise of Meta-Skills (Theme 3): Cognitive skills, like design/architectural thinking and prompt engineering, together with interpersonal skills like communication and collaboration, is becoming essential for working in complex, AI-enhanced environments.

Gaps in Organizational Support (Theme 4): Despite individual and even enthusiastic interest in AI, organizations often lack clear strategies, formal training, and policies.

This means that professionals have to deal with the changes imposed by AI on their own.

Tensions Around Career Progression (Theme 5): AI creates uncertainty, especially for junior roles. People worry about higher entry barriers and the potential loss of basic learning experiences. This could lead to a drop in overall skills, making them less superficial.

The Polarity of Perceived Risks and Opportunities (Theme 6): Professionals have mixed opinions about the future of AI and its impact on their jobs. Some view AI as a useful tool for advancing their careers, while others see it as a serious threat to job security and the depth of their skills.

[AI-assisted content generated with ChatGPT (OpenAI, 2025), reviewed by the author].

6.3 Answers to Research Questions

The study's findings directly address the research questions:

Main Question: How does AI adoption in software engineering influence required skillsets and reshape career progression? AI adoption without question fundamentally alters existing skills by automating routine technical tasks. This automation boosts demand for development of meta-skills like communication, as well as critical and architectural thinking. As a result, there are tensions in career progression. Junior roles face higher entry barriers and challenging learning paths, while on the other side, adaptable and capable professionals can find new, higher-leverage opportunities.

Supporting Question 1: Which technical skills are becoming less relevant, and which are emerging as essential? Skills such as programming language syntax memorization or repetitive template coding are losing relevance. On the other hand, emerging skills that are becoming essential now include evaluating AI outputs, prompt engineering, system design and structure, and improved collaboration abilities to work in cross-functional teams.

Supporting Question 2: How do software engineers perceive the impact of AI on career opportunities and risks? Perceptions vary. Many engineers view AI as a chance for faster skill development, higher productivity, and having a more significant strategic role within an organization. However, a considerable number of them also

see concerns, including higher entry barriers for junior roles, less foundational learning, and potential job loss, leading to uncertainty in career progression.

Supporting Question 3: What approaches are organizations taking to address these challenges? Currently, organizational approaches are mostly informal, reactive, and flawed. While some companies offer limited time for self-directed learning, there is a noticeable lack of structured AI training programs, clear policies, and strategic support initiatives even for companies claiming to be "AI First". This creates a significant gap between individual adaptation efforts and institutional assistance.

6.4 Theoretical Contributions

This study adds to several theoretical frameworks:

Dynamic Capabilities Framework - the findings show that many organizations do not have the dynamic abilities to sense the implications of AI or seize presenting opportunities through strategic investment in upskilling and transformation. The findings call for organizations to shift from reactive to proactive mode and build stronger dynamic capabilities in order to stay relevant and competitive.

Triple-Loop Learning Framework - the study reveals that while individual learning during AI adoption occurs at first and second-order level (task optimization and norm modification) many organizations are far from third-order learning since this level involves significant cultural and systemic change in response to AI.

Skill-Biased Technological Change - the research empirically supports SBTC in software engineering because it ascertains that AI tends to disproportionately favor professionals with advanced meta-skills while potentially reducing the number of entry-level positions, thus widening the skill gap.

6.5 Recommendations

Based on the study's findings, the following recommendations are offered for software engineering professionals and organizations:

6.5.1 For Software Engineering Professionals

Nurture Meta-Skills - actively work on developing critical thinking, problem framing, solution/architectural design and effective communication. It's certain that these skills will set human engineers apart in a world with AI.

Maintain Foundational Knowledge - do not use AI as a "black box". Keep strengthening your expertise in core engineering principles and always reflect on foundational knowledge to master your craft. This will help you in evaluating AI outputs critically and creating more refined solutions, as well as improving your problem-solving skills.

Embrace Proactive Learning - be proactive and take charge of your learning of AI tools and prompt engineering, as well as ethical AI principles. Formal support from organizations may not always be available or may be delayed.

Advocate for Organizational Support - be vocal and share your needs for structured training, access to AI tools and clear career paths with management. Join internal discussions to help organizations shape AI strategies.

6.5.2 For Organizations

Develop an AI Adoption Framework - go beyond current "informal" AI use by creating clear policies, governance frameworks, and security guidelines for AI tools. Detail how AI fits into the daily work of engineers and the software development process in general, outlining the changes and expectations.

Invest in Upskilling and Reskilling Programs - set up formal training programs that will cover new skills that are needed to ride the AI wave. These programs should go beyond passive learning time and cover mentorship, hands-on workshops, and should be included in the performance development and bonus plans of employees.

Redefine Roles and Career Paths - look into AI affects on different roles, especially pay attention on junior positions. Update job descriptions to highlight meta-skills and collaboration with AI. Work on creating clear career paths that include AI skills and offer strong mentorship to ensure a healthy talent pipeline and retention.

Foster a Culture of Continuous Learning and Ethical AI - include people with different skills in development teams to support teamwork across diverse fields. Foster a "learn-it-all" mindset and include check against ethical AI principles in every step of SDLC, from design to deployment. This approach ought to facilitate the ethical and pragmatic implementation of AI.

6.6 Conclusion

The adoption of AI has a significant and multi-layered impact on the software engineering profession, as this paper has shown through its previous chapters.

Results presented in this research support the fact that AI is a trigger for significant changes that will shape the software engineering industry in the years to come, especially in the necessary skill sets, career paths, and organisational dynamics, being far more important than just a tool for automation and abstraction of development work. Even though it is not questionable that AI has enormous potential for innovation and increased productivity, integrating it is not easy and comes with a number of difficulties like widespread lack of strategic organisational support, potential capability loss, and career uncertainty (with all the fear and stress it brings). In the AI era, the future of software engineering will depend on people's proactive adaptability and organisations' thoughtful, strategic responses to promote a cooperative, ethical, and ever-learning environment. Success for professionals and the sector will be determined by the capacity to maximise AI's potential while at the same time reducing its numerous risks.

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